CHAPTER 08

Time and Work

This chapter is one of the easiest chapters for the students. Even an average student can perform better than that in other chapters. There is basically one concept involved in this chapter *i.e.,* concept of efficiency.

So most of the problems are very similar in their basic characteristic. Almost every aptitude exam asks the problems from this chapter. On an average 2-3 problems from this chapter have been asked in past years in CAT.

As it is very clear to all of us that the work is directly related with time. As one can say if a particular person or machine works for more time then more work will be done and if it devotes less time then it yields less work *i.e.,* output of a machine or person is directly proportional to time, provided he/she maintains his/her efficiency during the work.

Chapter Checklist

- *Concept of Efficiency*
- *Relation Between Efficiency and Time*
- *Concept of Negative Work*
- *Applications of Inverse Proportion (or Product Constancy)*
- *Relation Between Efficiencies*
- ***CAT Test***

8.1 Concept of Efficiency

Suppose a person can complete a particular work in 2 days then we can say that each day he does half of the work or 50% work each day.

Thus it is clear that his efficiency is 50% per day. Efficiency is generally considered with respect to the time. The time can be calculated either in days, hours, minutes or months etc. So if a person completes his work in 4 days, then his efficiency (per day) is 25%.

Since each day he works 1/4th of the total work (*i.e.,* 25% of the total work). I would like to mention that the calculation of percentage and conversion of ratios and fractions into percentage and vice versa is the prerequisite for this chapter.

Now, if a person can complete a work in n days then his one day's work $= 1/n$ and this one day's work in terms of percentage is called his efficiency.

Also if a person can complete $\frac{1}{n}$ work in one day, then he can complete the whole work in n days.

Relation between work of 1 unit of time and percentage efficiency A person can complete his work in n days, then his one day's work $= 1/n$, his percentage efficiency $= 1/n \times 100$

No. of days/hours etc. required to complete the whole work	Work of 1 day/hour	Percentage efficiency
n	$1/n$	$100/n$
1	1/1	100%
2	1/2	50%
3	1/3	$33.33\% = 33\frac{1}{3}\%$
4	1/4	25%
5	1/5	20%
6	1/6	$16.66\% = 16\frac{2}{3}\%$
7	1/7	$14.28\% = 14\frac{2}{7}\%$
8	1/8	12.5%
9	1/9	$11.11\% = 11\frac{1}{9}\%$
10	1/10	10%

This table is very similar to the percentage fraction table given in the chapter of percentage. This table just manifests as a model for efficiency conversion.

Basically for faster and smarter calculation you have to have your percentage calculation very smart. All the problems of this chapter can be solved through two methods :

1. **Unitary method** 2. **Percentage efficiency**

Unitary method is generally obsolete in respect to high level aptitude exam of CAT since it involves difficult calculation of LCM each and every time. But when the problems are solved through percentage efficiency it becomes inevitable to save the time which in turn helps to do a few more problems within the stipulated time. I admit that initially it might be difficult to solve for those students who are not so good and confident in percentage and fraction calculation, but a little bit of extra Practice will yield an unexpected result in quicker calculations. Now I have some good examples to show you both the methods of solving the same problems.

Exp. 1) A can do a job in 12 days and B can do the same job in 6 days, in how many days working together they can complete the job?

Solution A's 1 day's work $= \frac{1}{12}$, B's 1 day's work $= \frac{1}{6}$

$\therefore (A + B)$'s 1 day's work s $= \frac{1}{12} + \frac{1}{6} = \frac{3}{12} = \frac{1}{4}$

$\therefore$ Time taken by both to finish the whole work

$$= \frac{1}{1/4} = 4 \text{ days}$$

Alternatively efficiency of A $= \frac{100}{12} = 8.33\%$

and efficiency of $B = \frac{100}{6} = 16.66\%$

Combined efficiency of A and B both $= 8.33 + 16.66 = 25\%$

$\therefore$ Time taken by both to finish the work (working together)

$$= \frac{100}{25} = 4 \text{ days}$$

NOTE As per my experience, I have found that in this chapter only selected (numerals) numbers are always used and thus there are almost 20 – 25 numbers are frequently used. So one can very easily remember (and calculate) the percentage efficiency but with different combinations to calculate the LCM becomes a very tedious job. Still you can choose your own method, which is comfortable to you.

Exp. 2) A can do a job in 10 days, B can do the same job in 12 days and C can do the same job in 15 days. In how many days they will finish the work together?

Solution A's 1 day's work $= \frac{1}{10}$, B's 1 day's work $= \frac{1}{12}$

C's 1 day's work $= 1 / 15$

$(A + B + C)$'s one day's work $= \frac{1}{10} + \frac{1}{12} + \frac{1}{15} = \frac{15}{60} = \frac{1}{4}$

Since they can complete $\frac{1}{4}$ work in 1 day. So they will finish the whole work in $\frac{1}{1/4} = 4$ days.

Alternatively A's efficiency $= 10\%$

B's efficiency $= 8.33\%$

C's efficiency $= 6.66\%$

Combined efficiency of A, B and $C = 10 + 8.33 + 6.66 = 25\%$

Hence, they will take $\frac{100}{25} = 4$ days to finish the job working together (Since in one day they complete 25% work)

8.2 Relation Between Efficiency and Time

Efficiency is inversely proportional to the time (*i.e.*, number of days, hours, minutes) etc.

For example if A is twice efficient as B, it means, A takes half the time to finish the same job as B requires working alone.

Exp. 1) A takes 16 days to finish a job alone, while B takes 8 days to finish the same job. What is the ratio of their efficiency and who is less efficient.

Solution Since A takes more time than B to finish the same job hence A is less efficient or efficiency of $A = \frac{100}{16} = 6.25\%$

and efficiency of $B = \frac{100}{8} = 12.5\%$

ratio of efficiency of $A : B = \frac{1}{16} : \frac{1}{8} = 1 : 2$

Hence, B is twice efficient as A.

Exp. 2) A is thrice efficient as B and A takes 20 days to do a job, then in how many days B can finish the same job?

Solution Ratio of efficiency of $A : B = 3 : 1$

$\therefore$ Ratio of required days of $A : B = \frac{1}{3} : \frac{1}{1} = 1 : 3$

Now since A takes 20 days. So B will take 60 days to finish the same job.

NOTE Ratio of number of days is equal to the ratio of the reciprocals of efficiency and *vice-versa.*

Exp. 3) P is thrice as efficient as Q and is therefore able to finish a piece of work in 60 days less than Q. Find the time in which P and Q can complete the work individually.

Solution Efficiency of $P : Q = 3 : 1$

Required number of days of $P : Q = 1 : 3$

i.e., if P requires x days then Q requires $3x$ days

but $3x - x = 60 \Rightarrow 2x = 60 \Rightarrow x = 30$ and $3x = 90$

Thus P can finish the work in 30 days and Q can finish the work in 90 days.

Exp. 4) A is twice as good a workman as B and is therefore able to finish a piece of work in 30 days less than B. In how many days they can complete the whole work; working together?

Solution Ratio of efficiency $= 2 : 1\ (A : B)$

Ratio of required time $= 1 : 2\ (A : B) \Rightarrow x : 2x$

but $2x - x = 30 \Rightarrow x = 30$ and $2x = 60$

now efficiency of $A = 3.33\%$ and efficiency of $B = 1.66\%$

Combined efficiency of A and B together $= 5\%$

$\therefore$ time required by A and B working together to finish the work

$$= \frac{100}{5} = 20 \text{ days.}$$

NOTE $\text{Efficiency} \propto \frac{1}{\text{number of time units}}$

$\therefore$ Efficiency $\times$ time $=$ constant work

Hence, Required time $= \frac{\text{work}}{\text{efficiency}}$

Whole work is always considered as 1, in terms of fraction and 100%, in terms of percentage.

In general, number of day's or hours $= \frac{100}{\text{efficiency}}$

Exp. 5) A can do a works in x days while B can do the same work in y days then in how many days will they complete the work, working together?

Solution A's one day's work $= \frac{1}{x}$, B's one day's work $= \frac{1}{y}$

So, both A and B completes $\frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}$ work in one day.

Now, **by Unitary method** $\frac{x+y}{xy}$ work can be completed in 1 day

$\therefore$ 1 (means complete) work will be finished in $\frac{1}{\frac{(x+y)}{xy}}$ days

$$= \frac{xy}{x+y} \text{ days}$$

$\left(\because \text{Required time to complete the work} = \frac{1}{\text{one day's work}}\right)$

8.3 Concept of Negative Work

In this case one person works but another destroyes it or cancels it. For example Sonu can write 20 pages per hour but his younger sister Rimjhim erases 10 pages per hour which Sonu writes. It means finally Sonu writes 10 pages per hour since each hour his sister erases 10 pages out of 20 written pages. Take an other example :

Exp. 1) A tub can be filled in 20 minutes but there is a leakage in it which can empty the full tub in 60 minutes. In how many minutes it can be filled?

Solution Filling efficiency $= 5\%$ $\left(\because 5 = \frac{100}{20}\right)$

emptying efficiency $= 1.66\%$ $\left(\because 1.66 = \frac{100}{60}\right)$

Net efficiency $= 5 - 1.66 = 3.33\%$

$\therefore$ Required time to fill the tub $= \frac{100}{3.33} = 30$ minutes

Alternatively In 1 minute tub is filling $= \frac{1}{20}$

In 1 minute tub is emptying $= \frac{1}{60}$

$\therefore$ In 1 minute, effective filling of tub $= \frac{1}{20} - \frac{1}{60} = \frac{2}{60} = \frac{1}{30}$

Since $\frac{1}{30}$ part of tub is filled in 1 minute.

Therefore complete tub will be filled in $\frac{1}{1/30} = 30$ minutes

8.4 Applications of Inverse Proportion (or Product Constancy)

As I have already discussed thoroughly the concept of inverse proportion and product constancy in ratio-proportion and it has been widely used in profit loss chapter also.

Since the efficiency or rate of work done in one unit of time (mentioned) is inversely proportional to the time *i.e.,* if the rate of work done is greater then the time required will be less and if the rate of work done is less then the time required for the same amount of work will be more. This product constancy method is limited to the constant work, if the amount of work gets changed, then it does not work, then we have to take help from the unitary method. When more than one man/machine work on a particular work/project, the rate of work is calculated as the strength of workers working in a particular time. So the amount of work is defined in terms of man-days or man-hours or man-days-hours.

Exp. 1) If 20 persons can do a piece of work in 7 days then calculate the number of persons required to complete the work in 28 days

Solution Number of persons × days = work

$20 \times 7 = 140$ man-days

Now, $x \times 28 = 140$ man-days $\Rightarrow x = 5$

Therefore in second case the required number of person is 5.

Second Method : Since work is constant, therefore

$$M_1 \times D_1 = M_2 \times D_2 = \text{ work done}$$

$$20 \times 7 = M_2 \times 28 \Rightarrow M_2 = 5$$

Third Method : Men × days = work, which is constant

$$\downarrow \frac{3}{4} \begin{pmatrix} 20 & & 7 \\ & & \\ 5 & & 28 \end{pmatrix} 3\uparrow$$

Since number of days is increased by 3 times (*i.e.*, 300%). So the number of men will be decreased by $\frac{3}{4}$ times (*i.e.*, 75%)

(remember percentage change graphic)

Exp. 2) If 25 men can do a piece of work in 36 days working 10 hours a day, then how many men are required to complete the work working 6 hours a day in 20 days?

Solution $M_1 \times D_1 \times H_1 = M_2 \times D_2 \times H_2$

$$25 \times 36 \times 10 = M_2 \times 20 \times 6 \Rightarrow M_2 = 75 \text{ persons}$$

Alternatively

Men	×	Time	
25		360	$\frac{2}{3}\downarrow$
75		120	

$2\uparrow$ [Since product (work) is constant]

By percentage change graphic, when time is decreased by $\frac{2}{3}$ (*i.e.*, 66.66%), number of men is increased by 2 times (*i.e.*, 200%)

Exp. 3) A contractor employed 30 men to complete the project in 100 days. But later on he realised that just after 25 days only 20% of the work had been completed.

(a) How many extra days, than the scheduled time are required?

(b) To complete the work on the scheduled time how many men he has to increase?

(c) If the amount of work is also increased by 20% of the actual work, then how many extra days are required (in comparison with scheduled time) but the number of men remained constant.

(d) How many men should be increased so that the work will be completed in 25 days less than the scheduled time.

Solution

(a) Men × days = work done

$30 \times 25 = 750 = 20\%$ of the actual work

Now, the work to be done is 4 times than the work done but the number of days is only 3 times. So he is required 4 times the number of days, thus he has to work for extra 25 days.

(b) $M_1 D_1 = M_2 D_2$, $4(30 \times 25) = M_2 \times 75 \Rightarrow M_2 = 40$

(Since, the work to be done is 4 times of the work done. Hence he requires 4 times man-days.)

Alternatively Since the new product (*i.e.*, work) is 4 times of the original product (*i.e.*, work). But the new product is being multiplied by 3. Thus to make it 4 times we have to multiply it by 4/3. Thus without changing number of days we get the new value of number of men which is 40 (being multiplied by 4/3) Therefore he has to increase 10 more men.

(c) New work = 3750 man-days and the available number of men = 30

$\therefore$ number of days required $= \frac{3750}{30} = 125$

So, he has to work for extra 50 days where $50 = (125 - 75)$

(d) Work = 3000 unit (man-days)

number of available days $= 50(75 - 25)$

$\therefore$ number of men required $= \frac{3000}{50} = 60$

Thus he has to increase $(60 - 30) = 30$ more men.

Exp. 4) 16 workers working 6 hours a day can build a wall of length 150 metres, breadth 20 m and height 12 m in 25 days. In how many days 12 workers, working 8 hours a day can build a wall of length 800 m, breadth 15 m and height 6 m.

Solution Here work is the volume of the wall. So the work force should be increased/decreased in the ratio of volume of the work. Therefore

$$\frac{L_1 B_1 H_1}{L_2 B_2 H_2} = \frac{M_1 D_1 T_1}{M_2 D_2 T_2}$$

where L, B, H are length, breadth and height of the wall respectively and M, D, T are men, days and time in hours per day, respectively. 1 indicates the first case, while 2 indicates second case. So the ratio of work force remains constant as the ratio of volume and work.

$$\therefore \quad \frac{150 \times 20 \times 12}{800 \times 15 \times 6} = \frac{16 \times 6 \times 25}{12 \times 8 \times D_2} \Rightarrow D_2 = 50$$

hence the required number of days = 50

Please notice that the volume of work becomes twice (in the second case) so the work force will also be twice to the previous work force.

8.5 Relation Between Efficiencies

In this case the efficiencies of different persons are different but when they work in a group, so the efficiency of the group is required to know the time taken. For example 3 men can do a work in 4 days while 12 boys can do the same work in 3 days. It means we need $3 \times 4 = 12$ man-days *i.e.*, 12 men can finish the job in 1 day.

Similarly we need $12 \times 3 = 36$ boys days *i.e.*, 36 boys can finish the same job in 1 day.

Here we can see that to finish the work in only 1 day 12 men are needed while 36 boys are needed. Thus we can conclude that work of 12 men is equal to the work of 36 boys. Therefore efficiency of 12 men is equal to 36 boys *i.e.*, we can say the efficiency of 1 man is equal to 3 boys.

Thus a man is thrice efficient as a boy or we can say that a man is two times more efficient than a boy.

Exp. 1) 6 boys and 8 women finish a job in 6 days and 14 boys and 10 women finish the same job in 4 days. In how many days working together 1 boy and 1 woman can finish the work?

Solution In this kind of questions we find the work force required to complete the work in 1 day (or given unit of time) then we equate the work force to find the relationship between the efficiencies (or work rate) between the different workers.

Therefore $6B + 8W = 6$ days

$\Rightarrow$ $6(6B + 8W) = 1$ day (inversely proportional)

$\Rightarrow$ $36B + 48W = 1$ (by unitary method)

Again $14B + 10W = 4$ days

$\Rightarrow$ $56B + 40W = 1$

So, here it is clear that either we employ $36B$ and $48W$ to finish the work in 1 day or $56B$ and $40W$ to finish the same job in 1 day. Thus, we can say

$\Rightarrow$ $36B + 48W = 56B + 40W$

$\Rightarrow$ $20B = 8W$

$\Rightarrow$ $W = 2.5B$

Thus a woman is 2.5 times as efficient as a boy.

Now, since $36B + 48W = 1$

$\Rightarrow$ $36B + 48 \times (2.5B) = 1$

$\Rightarrow$ $156B = 1$

i.e., to finish the job in 1 day 156 boys are required or the amount of work is 156 boys-days.

Again $1W + 1B = 2.5B + 1B = 3.5B$

Now, since 156 boys can finish the job in 1 day

So 1 boy can finish the job in 1×156 days

$\therefore$ 3.5 boys can finish the job in

$$\frac{1 \times 156}{3.5} = 44\frac{4}{7} \text{ days}$$

NOTE There is a great difference between 'and' & 'or' For example 4 men and 8 women can do a piece of work in 10 days : means it is unknown that who is faster or slower (*i.e.*, we don't know the relation between efficiencies of a man and a woman). Again, 4 men or 8 women can do a piece of work in 10 days : means 4 men are equal to 8 women. Hence a man is twice efficient as a woman.

Exp. 2) 6 men and 8 women can do a job in 10 days. In how many days can 5 men and 9 women do the same job?

Solution Since we don't know the relation between work rate (or efficiency) between a man and that of a woman so we can't find the required number of days.

Exp. 3) 6 men and 8 women can do a job in 10 days. In how many days can 3 men and 4 women finish the same job working together?

Solution Notice here we don't know the relation of efficiencies but we can solve the problem due to clear relation between the work force.

Since $6M + 8W = 10$ days

$2(3M + 4W) = 10$ days

$\Rightarrow$ $(3M + 4W) = 20$ days

Since, the work force has become half of the original force so number of days must be double.

Thus required number of days = 20

Exp. 4) A can complete a work in 12 days, B in 15 days. Find the time taken by them :

(a) when A and B worked together.
(b) when A and B worked alternatively started by A.
(c) when A and B worked alternatively started by B.
(d) if A started two days later, in comparison to B.
(e) if B started two days later, in comparison to A.
(f) if A leaves two days before the actual completion of the work.
(g) if B leaves two days before the actual completion of the work.
(h) if A leaves two days before the scheduled completion of the work.
(i) if B leaves two days before the scheduled completion of the work.
(j) if B does negative work with his same work rate.

Solution (a) A's efficiency = 8.33%

B's efficiency = 6.66%

Combined efficiency = 15%

So, the required time $= \frac{100}{15} = 6\frac{2}{3}$ days.

(b) In every two days A and B work 15%. So in 12 days they will complete 90% work. Now on the 13th day, A will finish 8.33% of the remaining (*i.e.*, 10% work) and the rest 1.66% will be finished by B on 14th day by taking time $= \frac{1.66}{6.66} = \frac{1}{4}$ day.

Thus, total required time $= 12 + 1 + \frac{1}{4} = 13\frac{1}{4}$ days.

See the chart below :

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Work	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	1.66
	15		15		15		15		15		15			
	$15 \times 6 = 90\%$												10%	

(c) Here the difference is in the last day/days only.

See the chart below :

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Work	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	8.33	6.66	3.33
	15		15		15		15		15		15		10	
	$15 \times 6 = 90\%$												10%	

So, after the completion of 90% work in 12 days, on the 13th day 6.66% work will be finished by B and then 3.33% (the last part of the work) will be finished by A by taking time $= \frac{3.33}{8.33} = \frac{2}{5}$ day.

Thus, the required time $= 12 + 1 + \frac{2}{5} = 13\frac{2}{5}$ days.

(d) It means B worked for 2 days more than A has worked for. Now, since B completes $2 \times 6.66 = 13.33\%$ work in two days. So the remaining work will be done by A and B together.

Therefore the time taken by A and B, working together

$$= \frac{86.66}{15}\frac{\text{(remaining work)}}{\text{(combined efficiency)}} = 5\frac{7}{9} \text{ days}$$

Thus the total required time to finish the whole work

$$= 2 + 5\frac{7}{9} = 7\frac{7}{9} \text{ days.}$$

(e) It means A alone worked for 2 days which is equal to $8.33 \times 2 = 16.66\%$

So, the remaining (83.33%) work will be done by A and B together.

$\therefore$ Time taken together $= \frac{83.33}{15} = 5\frac{5}{9}$ days

Thus, the total required time $= 2 + 5\frac{5}{9} = 7\frac{5}{9}$ days.

(f) 'A leaves two days before the actual completion of the work' means in the last two days B has worked alone which is $6.66 \times 2 = 13.33\%$

Therefore the remaining work *i.e.*, 86.66% work was only done by A and B together in $\frac{86.66}{15} = 5\frac{7}{9}$ days

Thus, the required time $= 2 + 5\frac{7}{9} = 7\frac{7}{9}$ days.

NOTE This case is exactly same as part (d) of the same example. The difference is only in order of days. In part (d) B worked alone initially for 2 days and in part (f) B worked alone in the last 2 days.

(g) Applying the same concept as above we can calculate the required number of days which is same as in case (e)

Work	$A + B$	A
	$15 \times 5\frac{2}{5}$	8.33×2
	83.33%	16.66%

(h) 'A leaves two days before the scheduled completion of the work' means A works with B only for $4\frac{2}{3}$ days since the scheduled time is $6\frac{2}{3}$ [see the case (a)]

Scheduled time is the time in which work could be finished by working together without any change in efficiency or work force. Now since both A and B have worked together for $4\frac{2}{3}$ days it means the last 2 days work (scheduled) which is equal to 30% (since if A and B working together can finish 30% work in 2 days) will be done by B only.

$\therefore$ required time to complete 30% work by B alone

$$= \frac{30}{6.66} = 4.5 \text{ days}$$

Thus the total required time to finish the whole work

$$= 4\frac{2}{3} + 4\frac{1}{2} = 9\frac{1}{6} \text{ days.}$$

(i) Very similar to the case (h) A and B worked only for $4\frac{2}{3}$ days (2 days less than the scheduled time $6\frac{2}{3}$ days)

(It is obvious that in $4\frac{2}{3}$ days A and B, working together completes 70% work, but we can directly calculate that they can complete 30% if they work together, in 2 days)

Thus 30% work is done by A alone

$\therefore$ time taken $= \frac{30}{8.33} = 3\frac{3}{5}$ days

Thus, the total time $= 4\frac{2}{3} + 3\frac{3}{5} = 8\frac{4}{15}$ days.

(j) Combined efficiency of A and $B = (8.33) + (-6.66)$

$= 1.66\%$

$\therefore$ Time required by A and B working together

$$= \frac{100}{1.66} = 60 \text{ days}$$

(Negative work means, B works against A)

Exp. 5) A can complete a work in 10 days, B in 12 days and C in 15 days. All of them began the work together, but A had to leave the work after 2 days of the start and B 3 days before the completion of the work. How long did the work last?

Solution See the diagram and then interpret the language of the question.

Initial 2 days		Last 3 days
$A + B + C$	$B + C$	C
$25 \times 2 = 50\%$	$15\% \times 2$ $[100 - (50 + 20)]$	$6.66 \times 3 = 20\%$

Since initially for 2 days all of them A, B and C work together so they complete the 50% work and for the last 3 days only C works which is equal to 20% work.

Thus, the remaining work $= 30\%[100 - (50 + 20)]$

This 30% work was done by B and C in 2 days $= \left(\frac{30}{15}\right)$.

NOTE

Efficiency of $A = 10\%$
Efficiency of $B = 8.33\%$
Efficiency of $C = 6.66\%$

So, the total number of required days $= 2 + 2 + 3 = 7$ days.

CAT-Test

Questions Helping you bell the CAT

LEVEL 01 > BASIC LEVEL EXERCISE

1 A can finish a piece of work in 12 days while B can do it in 15 days. If both work at it together, what time will they take to do the work?
(a) 6 days (b) 8 days
(c) $6\frac{2}{3}$ days (d) $10\frac{1}{5}$ days

2 A can do a piece of work in 10 days, B can do it in 12 days and C can do it in 15 days. In how many days can they complete the work, working together?
(a) 6 (b) 4 (c) 5 (d) 10

3 A can do a piece of work in 5 days, B can do it in 10 days. With the help of C, they finish the work in 2 days. In how many days can C alone do the whole work?
(a) 3 (b) 4 (c) 5 (d) 6

4 A can do a piece of work in 8 days, B can do it in 16 days, while C can do it in 80 days. In how many days can they complete the whole work, working together?
(a) 5 (b) 6 (c) $8\frac{2}{3}$ (d) $20\frac{2}{5}$

5 A can do a piece of work in 10 days, B can do it in 15 days working together they can finish the work in :
(a) 9 (b) 8 (c) 10 (d) 6

6 A can do a piece of work in 24 days, while B can do it in 30 days. With the help of C they can finish the whole work in 12 days. How much time is required for C to complete the work, alone?
(a) 100 days (b) 120 days
(c) 125 days (d) 72 days

7 A can do a piece of work in 10 days. B can do it in 24 days. If C also works with them then it takes only 6 days to complete the whole work. In how many days can C alone complete the whole work?
(a) 25 (b) 40 (c) 50 (d) 75

8 A can do a piece of work in 14 days while B can do it in 21 days. In how many days, working together will they complete the whole work?
(a) 10.5 (b) 8 (c) 8.4 (d) 9

9 A can do a piece of work in 24 days. If B is 60% more efficient than A, then the number of days required by B to do the same piece of work is :
(a) 10 (b) 15 (c) 12 (d) 9.6

10 A is twice as good a workman as B and together they finish a piece of work in 14 days. In how many days can A alone finish the work?
(a) 18 (b) 20 (c) 21 (d) 24

11 A is twice as good a workman as B and together they finish a piece of work in 7 days. In how many days can A alone finish the work?
(a) 10 (b) 10.5 (c) 12 (d) 11

12 A is thrice as good a workman as B and takes 10 days less to do a piece of work than B takes. The number of days taken by B to finish the work is :
(a) 9 (b) 18 (c) 15 (d) 12

13 A is twice as good a workman as B and therefore A takes 6 days less than B to finish the work individually. If A and B working together complete the work in 4 days, then how many days are required by B to complete the work alone?
(a) 12 (b) 18 (c) 8 (d) 6

14 A is thrice efficient as B and C is twice as efficient as B. What is the ratio of number of days taken by A, B and C, when they work individually?
(a) 2 : 6 : 3 (b) 2 : 3 : 6 (c) 1 : 2 : 3 (d) 3 : 1 : 2

15 Ajit is 3 times as efficient as Bablu, then the ratio of number of days required by each to work alone, completely?
(a) 3 : 1 (b) 1 : 3
(c) 6 : 3 (d) 3 : 6

16 X is 4 times more efficient as Y, then the number of days taken by X and Y are in the ratio :
(a) 8 : 1 (b) 1 : 5 (c) 6 : 8 (d) 5 : 1

17 If A takes 4/5 days as B takes and working together they require $\frac{20}{3}$ days to complete the whole work. What is the efficiency of B ?
(a) $6\frac{2}{3}\%$ (b) 16% (c) 5.55% (d) $8\frac{1}{3}\%$

18 A is thrice as efficient as B. Working together they complete the work in 3 days. If B takes 8 days more than A, what is the number of days taken by A to finish the whole work, alone?
(a) 4 (b) 2
(c) 12 (d) 16

19 A and B can do a piece of work in 8 days, B and C can do the same work in 12 days and A and C complete it in 8 days. In how many days can A, B and C complete the whole work, working together?
(a) 4 (b) 6
(c) 12 (d) 9

20 A and B can do a piece of work in 12 days, B and C in 15 days C and A in 20 days. In how many days can C alone do it?
(a) 60 (b) 50
(c) 25 (d) 24

21 Ganga, Jamuna and Saraswati can do a piece of work, working together, in 1 day. Ganga is thrice efficient as Jamuna and Jamuna takes the twice the number of days as Saraswati takes to do it alone. What is the difference between the number of days taken by Ganga and Saraswati?
(a) 1 (b) 2
(c) 3 (d) 4

22 A can finish a work in 12 days and B can do it in 15 days. After A had worked for 3 days, B also joined A to finish the remaining work. In how many days, the remaining work will be finished?
(a) 3 (b) 4
(c) 5 (d) 6

23 Raja can do a piece of work in 14 days, while Rani can do the same work in 21 days. They started the work together but 3 days before the completion of the work, Raja left the work. The total number of days to complete the work is :
(a) 7 (b) 8.5
(c) 5 (d) $7\frac{1}{5}$

24 A and B can complete a task in 30 days when working together after A and B have been working together for 11 days, B is called away and A, all by himself completes the task in the next 28 days. Had A been working alone, the number of days taken by him to complete the task would have been :
(a) $33\frac{3}{19}$ (b) $19\frac{6}{25}$
(c) $44\frac{4}{19}$ (d) none of these

25 Sonu can do a piece of work in 20 days. He started the work and left after some days, when 25% work was done. After it Abhijeet joined and completed it working for 10 days. In how many days Sonu and Abhijeet can do the complete work, working together?
(a) 6 (b) 8
(c) 10 (d) 12

26 Efficiency of Asha is 25% more than Usha and Usha takes 25 days to complete a piece of work. Asha started a work alone and then usha joined her 5 days before actual completion of the work. For how many days did Asha work alone?
(a) 9 (b) 11 (c) 10 (d) 15

27 Krishna can do a work in 10 days while Mohan can do the same work in 20 days. They started work together. After 3 days Krishna left the work and Mohan completed it. For how many days Mohan worked alone more than the number of days required when both worked together?
(a) $4\frac{1}{3}$ (b) $3\frac{1}{4}$ (c) $2\frac{3}{5}$ (d) $3\frac{2}{3}$

28 Kareena can do a piece of work in 9 days and Karishma can do the same work in 18 days. They started the work. After 3 days Shahid joined them, who can complete alone the same whole work in 3 days. What is the total number of days in which they had completed the work?
(a) 12 (b) 8
(c) 4 (d) 6

29 Kavita, Babita and Samita started a work. 5 days later Samita left the work and Babita left the work after working 8 days. In how many more days Kavita would have completed the rest work if they take 20, 60 and 30 days individually to finish a work?
(a) 4 (b) 5 (c) 6 (d) 8

30 The ratio of efficiency of A is to C is 5 : 3. The ratio of number of days taken by B is to C is 2 : 3. A takes 6 days less than C, when A and C completes the work individually. B and C started the work and left after 2 days. The number of days taken by A to finish the remaining work is :
(a) 4.5 (b) 5 (c) 6 (d) $9\frac{1}{3}$

31 Anand can do a piece of work in 45 days, but Bahuguna can do the same work in 5 days less, than Anand, when working alone. Anand and Bahuguna both started the work together but Bahuguna left after some days and Anand finished the remaining work in 56 days with half of his efficiency but he did the work with Bahuguna with his complete efficiency. For how many days did they work together?
(a) 6 (b) 8
(c) 9 (d) 12

32 Chandni and Divakar can do a piece of work in 9 days and 12 days respectively. If they work for a day alternatively, beginning with Chandni, then in how many days will the work gets completed ?
(a) $10\frac{1}{4}$ (b) $9\frac{1}{5}$ (c) 11.11 (d) 10

33 Fatima and Zahira can do a piece of work in 12 days and 15 days respectively. If they work for alternate day and Fatima starts the work first, then in how many days will the work gets completed ?
(a) $12\frac{1}{5}$ (b) $13\frac{1}{4}$ (c) $13\frac{1}{5}$ (d) 15

34 In the previous question if Zahira starts first, then in how many days will the work gets completed ?
(a) $14\frac{1}{5}$ (b) 14
(c) $13\frac{1}{5}$ (d) $13\frac{2}{5}$

35 The number of days required by A, B and C to work individually is 6, 12 and 8 respectively. They started a work doing it alternatively. If A has started then followed by B and so on, how many days are needed to complete the whole work?
(a) 8 (b) 7.5 (c) 8.5 (d) $9\frac{1}{2}$

36 In the previous question if the order of working days be as B, C, A, B, C, A … (starting with B and followed by C and A respectively), then in how many days the work will be completed?
(a) $7\frac{3}{4}$ (b) 8 (c) $8\frac{1}{4}$ (d) 9

37 A takes 6 days less than B to do a certain job and 2 days more than C. A and B together can do the work in the same time as C. In how many days B alone can do the complete work?
(a) 10 (b) 14 (c) 12 (d) 16

38 A and B undertook a work for ₹ 350. A got ₹ 150 more than that of B, when they worked together. B takes 9 days more than A, when they work individually. In how many days A and B working together can do the whole work :
(a) 5 (b) $4\frac{2}{7}$ (c) $4\frac{5}{7}$ (d) $5\frac{4}{7}$

39 Alen and Border can do a work individually in 21 and 42 days respectively. In how many days they can complete the work, working alternatively?
(a) 14 (b) 28 (c) 42 (d) 35

40 C takes twice the number of days to do a piece of work than A takes. A and B together can do it in 6 days while B and C can do it in 10 days. In how many days A alone can do the work?
(a) 60 (b) 30
(c) 6 (d) 7.5

41 When A, B and C are deployed for a task, A and B together do 70% of the work and B and C together do 50% of the work. Who is most efficient?
(a) A (b) B
(c) C (d) can't be determined

42 Colonel, Major and General started a work together for ₹ 816. Colonel and Major did $\frac{8}{17}$ of the total work, while Major and General together did $\frac{12}{17}$ of the whole work. What is the amount of the least efficient person?
(a) ₹ 256 (b) ₹ 144
(c) ₹ 85 (d) can't be determined

43 Sharma is 20% less efficient than Kelkar. If Kelkar can do a piece of work in 24 days. The number of days required by Sharma to complete the same work alone?
(a) 20 (b) 30
(c) 28.8 (d) can't be determined

44 30 persons can do a piece of work in 24 days. How many more people are required to complete the work in 20 days?
(a) 4 (b) 5
(c) 6 (d) none of these

45 12 women can do a piece of work in 20 days. If the 4 women deny to work, then how many more days are required?
(a) 6 (b) 10
(c) 15 (d) none of these

46 35 boys can do a piece of work in 15 days. The work was completed in 25 days. How many boys did not turn up for the job?
(a) 14 (b) 20
(c) 6 (d) 7

47 24 men can complete a job in 40 days. The number of men required to complete the job in 32 days is :
(a) 30 (b) 40
(c) 25 (d) 50

48 16 men finished one-third work in 6 days. The number of additional men are required to complete the job in next 6 days :
(a) 10 (b) 8
(c) 16 (d) 32

49 If 10 persons can do a job in 20 days, then 20 person with twice the efficiency can do the same job in :
(a) 5 days (b) 40 days
(c) 10 days (d) 20 days

50 A certain job was assigned to a group of men to do in 20 days. But 12 men did not turn up for the job and the remaining men did the job in 32 days. The original number of men in the group was :
(a) 32 (b) 36
(c) 42 (d) 40

51 30 workers can finish a work in 20 days. After how many days should 9 workers leave the job so that the work is completed in total 26 days :
(a) 12 (b) 10
(c) 6 (d) none of these

52 25 men can complete a job in 30 days. After how many days should the strength of work force be increased by 50 men so that the work will be completed in $\frac{2}{3}$rd of the actual time:
(a) 15 (b) 10
(c) 18 (d) 5

53 A group of workers can complete a job in 120 days. If there were 4 more such workers then the work could be finished in 12 days less. What was the actual strength of workers?
(a) 30 workers (b) 40 workers
(c) 42 workers (d) 36 workers

54 Mr. Modi can copy 40 pages in 10 minutes, Mr Xerox and Mr. Modi both working together can copy 250 in 25 minutes. In how many minutes Mr. Xerox can copy 36 pages?
(a) 5 minutes (b) 6 minutes
(c) 3 minutes (d) 12 minutes

55 20 persons completed $\frac{1}{3}$rd of the work in 12 days. How many more person are required to finish the rest work in next 12 days?
(a) 20 (b) 12 (c) 18 (d) 40

56 A contractor undertook a work to complete in 60 days. But just after 20 days he observed that only $\frac{1}{5}$th of the project work had been completed. To complete the work in time (*i.e.,* in rest days) minimum how many workers he had to increase, if there were initially 75 workers were deployed for the task?
(a) 25 (b) 50
(c) 75 (d) can't be determined

57 6 men or 10 women can reap a field in 15 days, then the number of days that 12 men and 5 women will take to reap the same field is :
(a) 5 (b) 6 (c) 8 (d) 12

58 If 2 men or 3 women or 4 boys can do a piece of work in 52 days, then the same piece of work will be done by 1 man, 1 woman and 1 boy in :
(a) 48 days (b) 36 days
(c) 45 days (d) none of these

59 2 men or 5 women or 7 boys can finish a work in 469 days, then the number of days taken by 7 men, 5 women and 2 boys to finish the work is :
(a) 134 (b) 106
(c) 100 (d) 98

60 6 children and 2 men complete a certain piece of work in 6 days. Each child takes twice the time taken by a man to finish the work. In how many days will 5 men finish the same work?
(a) 6 (b) 8
(c) 9 (d) 15

61 2 men and 3 women finish 25% of the work in 4 days, while 6 men and 14 women can finish the whole work in 5 days. In how many days will 20 women finish it?
(a) 20 (b) 25
(c) 24 (d) 88

Directions (for Q. Nos. 62 and 63) *A can do a work in 15 days and B can do it in 18 days. With the help of C, all of them complete the work in 6 days.*

62 How long will it take *C* to finish the work alone?
(a) 30 days (b) 22 days
(c) $\frac{45}{2}$ days (d) 25 days

63 *A*, *B* and *C* received total ₹ 27,000 for the whole work. What is the share of *C*, if the money is distributed in the ratio of amount of work done, individually?
(a) 2700 (b) 7200
(c) 14400 (d) 6300

Directions : *Solve the following questions individually.*

64 314 weavers weaves 6594 shawls in 1 / 6 hours. What is the number of shawls weaved per hour by an average weaver?
(a) 42 (b) 21 (c) 102 (d) 126

65 Three men and two women can do a piece of work in 4 days, while two men and three women can do the same work in 5 days. ₹ 44 is given to a woman for her contribution towards work, per day. What is the amount received by a man per day?
(a) ₹ 88 (b) ₹ 144
(c) ₹ 154 (d) can't be determined

66 30 girls can finish a work in 40 days. After how many days should 10 girls leave the work so that it may be finished in 46 days, if all the 30 girls started the work?
(a) 18 days (b) 28 days
(c) 22 days (d) 30 days

67 If 8 women collect 200 kg of tea leaves in 10 hours. How many more (in kg) of tea leaves will 12 women collect in 8 hours?
(a) 24 kg (b) 40 kg
(c) 50 kg (d) 100 kg

68 450 man-days of work can be completed by a certain number of men in some days. If the number of people (men) are increased by 27, then the number of day required to complete the same work is decreased by 15. The number of days are required to complete the three times work (than the previous/actual work) by 27 men?
(a) 50 (b) 60 (c) 54 (d) 45

69 4 boys and 5 girls can do a piece of work in 10 days. 6 boys and 6 girls can do the same work in 7 days. In how many days can 2 boys and 7 girls complete the same work, working together?
(a) 15 days (b) 14 days
(c) 21 days (d) 18 days

70 If 20 engineers and 20 workers can together construct a 20 km road in 20 days. 40 engineers and 40 workers together construct 40 km road in how many days?
(a) 10 (b) 20
(c) 40 (d) can't be determined

71 $(x - 2)$ men can do a piece of work in x days and $(x + 7)$ men can do 75% of the same work in $(x - 10)$ days. Then in how many days can $(x + 10)$ men finish the work?
(a) 27 days (b) 12 days
(c) 25 days (d) 18 days

72 A man, a woman and a girl worked for a contractor for the same period. A man is twice efficient as a woman and a woman is thrice efficient as a girl ₹ 10000 were given to all of them. What is the sum of money received by a woman and a girl together?
(a) ₹ 5500 (b) ₹ 4500
(c) ₹ 4000 (d) ₹ 6000

73 33 men can do a job in 30 days. If 44 men started the job together and after every day of the work, one person leaves. What is the minimum number of days required to complete the whole work?
(a) 21 (b) 42
(c) 45 (d) none of these

74 Abhishek can do a piece of work in 40 days. He alone worked at it for 8 days and then Bacchhan completed alone the rest work in 24 days. In how many days they will complete the whole work, working together?
(a) $17\frac{1}{7}$ days (b) $18\frac{1}{7}$ days
(c) $9\frac{1}{6}$ days (d) 14 days

75 *C* is twice efficient as *A*. *B* takes thrice as many days as *C*. *A* takes 12 days to finish the work alone. If they work in pairs (*i.e., AB*, *BC*, *CA*) starting with *AB* on the first day then *BC* on the second day and *AC* on the third day and so on, then how many days are required to finish the work?
(a) $6\frac{1}{5}$ days (b) 4.5 days
(c) $5\frac{1}{9}$ days (d) 8 days

76 Ahluwalia and Bimal together take 6 days to finish the work. Bimal and Jalan together take 10 days to finish the work. What is the difference between number of days taken by Ahluwalia and Jalan when they worked alone to complete the whole work?
(a) 12 days
(b) 16 days
(c) 15 days
(d) can't be determined

77 *B* is twice efficient as *A* and *A* can do a piece of work in 15 days. *A* started the work and after a few days *B* joined him. They completed the work in 11 days, from the starting. For how many days they worked together?
(a) 1 day (b) 2 days
(c) 6 days (d) 5 days

78 *A*, *B* and *C* can complete a piece of work in 15, 30 and 40 days respectively. They started the work together and *A* left 2 days before the completion of the work and *B* left 4 days before the completion of the work. In how many days was the work completed?
(a) $7\frac{3}{10}$ (b) $10\frac{2}{15}$
(c) $10\frac{7}{30}$ (d) none of these

79 There was a leakage in the container of the refined oil. If 11 kg oil is leaked out per day then it would have lasted for 50 days, if the leakage was 15 kg per day, then it would have lasted for only 45 days. For how many days would the oil have lasted, if there was no leakage and it was completely used for eating purpose?
(a) 80 days (b) 72 days
(c) 100 days (d) 120 days

80 A contractor undertook to complete the work in 40 days and he deployed 20 men for his work. 8 days before the scheduled time he realised that $\frac{1}{3}$rd of the work was still to be done. How many more men were required to complete the work in stipulated time?
(a) 16 (b) 15 (c) 20 (d) 25

81 7 Indian and 4 Chinese finish a job in 5 days. 7 Japanese and 3 Chinese finish the same job in 7 days. Given that the efficiency of each person of a particular nationality is same but different from others. One Indian, one Chinese and one Japanese will complete the work in :
(a) $18\frac{3}{13}$ days (b) $20\frac{5}{12}$ days
(c) $21\frac{6}{14}$ days (d) $20\frac{7}{12}$ days

82 *A*, *B* and *C* are three book binders. *A* takes 8 minutes, *B* takes 12 minutes and *C* takes 16 minutes to bind a book. If they work each day for 12 hours, then on an average, how many books each one bind per day?
(a) 65 (b) 52 (c) 48 (d) 70

83 A piece of work can be completed by 10 men and 6 women in 18 days. Men works 9 hours per day while women works 7.5 hours per day. Per hour efficiency of a woman is 2/ 3rd of a man's efficiency. In how many days 10 men and 9 women complete the work?
(a) 16 days (b) 20 days (c) 30 days (d) 25 days

84 *B* and *C* are equally efficient, but the efficiency of *A* is half of each *B* and *C*. *A* and *B* started a work and 3 days later *C* joined them. If *A* alone can do the work in 14 days, then in how many more days the work will be completed?
(a) 1 (b) 2 (c) 3 (d) 4.5

85 *A* can do a piece of work in 10 days, *B* in 15 days. They work together for 5 days, the rest of the work is finished by *C* in two more days. If they get ₹ 3000 as wages for the whole work, what are the daily wages of *A*, *B* and *C* respectively (in ₹) :
(a) 200,250,300 (b) 300,200,250
(c) 200,300,400 (d) none of these

86 *A* can do a piece of work in 2 hours, *B* can do thrice the work in 8 hours and 'C' can do the same work as *A* in 8 hours. If all of them work together, how long it would take them to complete the work :
(a) 1 hour (b) 2 hours
(c) 3 hours (d) 4 hours

87 A is twice efficient as B and together they do the same work in as much time as C and D together. If C and D can complete the work in 20 and 30 days respectively, working alone, then in how many days A can complete the work individually :
(a) 12 days (b) 18 days (c) 24 days (d) 30 days

88 4 men and 2 boys can finish a piece of work in 5 days. 3 women and 4 boys can finish the same work in 5 days. Also 2 men and 3 women can finish the same work in 5 days. In how many days 1 man, 1 woman and one boy can finish the work, at their double efficiency?
(a) $4\frac{8}{13}$ (b) $4\frac{7}{13}$
(c) $3\frac{7}{13}$ (d) none of these

89 If m men can do a work in r days, then the number of days taken by $(m + n)$ men to do it is :
(a) $\frac{m+n}{mn}$ (b) $\frac{m+n}{mr}$ (c) $\frac{mr}{(m+n)}$ (d) $\frac{(m+n)r}{mn}$

90 Pipe A can fill a tank in 36 minutes and pipe B can fill it in 45 minutes. If both the pipes are opened to fill an empty tank, in how many minutes will it be full?
(a) 15 (b) 18 (c) 20 (d) 25

91 Tap A can fill the empty tank in 12 hours, but due to a leak in the bottom it is filled in 15 hours. If the tank is full and then tap A is closed then in how many hours the leak can empty it?
(a) 45 hours (b) 48 hours
(c) 52 hours (d) 60 hours

92 Pipe A and B can fill a cistern in 10 hours and 15 hours respectively. When a third pipe C which works as an outlet pipe is also open then the cistern can be filled in 18 hours. The outlet pipe can empty a full cistern in :
(a) 12 hours (b) 8 hours
(c) 9 hours (d) 14 hours

93 A cistern has a leak which would empty it in 6 hours. A tap is turned on which fills the cistern @ 10 liters per hour and then it is emptied in 15 hours. What is the capacity of the cistern?
(a) 100 litres (b) 166.66 litres
(c) 60.66 litres (d) none of these

94 Tap a fills a tank in 10 hours and B can fill it in 15 hours. Both are opened simultaneously. Sometimes later tap B was closed, then it takes total 8 hours to fill up the whole tank. After how many hours B was closed?
(a) 2 (b) 3
(c) 4 (d) 5

95 Tap A can fill a tank in 20 hours, B in 25 hours but tap C can empty a full tank in 30 hours. Starting with A, followed by B and C each tap opens alternatively for one hour period till the tank gets filled up completely. In how many hour the tank will be filled up completely?
(a) $51\frac{11}{15}$ (b) $52\frac{2}{3}$
(c) $24\frac{4}{11}$ (d) none of these

96 If one pipe A can fill a tank in 20 minutes, then 5 pipes, each of 20% efficiency of A, can fill the tank in :
(a) 80 min (b) 100 min
(c) 20 min (d) 25 min

97 Pipe A basically used as inlet pipe and pipe B is used as outlet pipe. Pipes A and B both are opened simultaneously, all the time. When pipe A fills the tank and B empty the tank, it will take double the time than when both the pipe fill the tank. When pipe B is used for filling the tank, its efficiency remains constant. What is the ratio of efficiency of pipe A and pipe B respectively?
(a) 3 : 1 (b) 5 : 2
(c) 1 : 3 (d) 3 : 2

98 Pipe A can fill the tank in 4 hours, while pipe B can fill it in 6 hours working separately. Pipe C can empty whole the tank in 4 hours. He opened the pipe A and B simultaneously to fill the empty tank. He wanted to adjust his alarm so that he could open the pipe C when it was half-filled, but he mistakenely adjusted his alarm at a time when his tank would be 3/4th filled. What is the time difference between both the cases, to fill the tank fully
(a) 48 min (b) 54 min
(c) 30 min (d) none of these

99 Two pipes A and B can fill a cistern in 15 hours and 10 hours respectively. A tap C can empty the full cistern in 30 hours. All the three taps were open for 2 hours, when it was remembered that the emptying tap had been left open. It was then closed. How many hours more would it take for the cistern to be filled?
(a) 30 min (b) 1.2 hours
(c) 24 min (d) 35 min

100 Pipe A can fill an empty tank in 30 hours while B can fill it in 45 hours. Pipe A and B are opened and closed alternatively *i.e.,* first pipe A is opened, then B, again A and then B and so on for 1 hour each time without any time lapse. In how many hours the tank will be filled when it was empty, initially?
(a) 36 (b) 54 (c) 48 (d) 60

Directions (for Q. Nos. 12 to 15) *Ready Tailoring Services is very well known in its quality and time bound services. The company (Ready Tailoring Services) received a large order for stiching military uniforms. It has two different orders to prepare the shirts one for Officers and second for Jawans (non-officers). It has three cutters who will cut the fabric. Six tailors who will do the stiching and 3 assistant to stich the buttons and iron the shirts. Each of these 12 persons will work for exactly 8 hours a day. Each of the Officers uniform requires 20 minutes for cutting the fabric, 1 hour for stiching and 20 minutes for stiching buttons and ironing the shirts. Whereas the Jawan's uniform requires 15 minutes for cutting the fabric and 60 minutes for stiching and 10 minutes for buttons and ironing.*

12 If the company has to supply 40 officers uniforms only and no other on a particular day, how many man-hours are utilised on that day?
(a) $33\frac{1}{3}$ hours (b) $66\frac{2}{3}$ hours
(c) 40 hours (d) 60 hours

13 If the number of tailors will be increased by 50% then maximum how many uniforms for officers can be completed in one day?
(a) 48 (b) 50
(c) 60 (d) 72

14 If the company can increase maximum 3 employees of any category then for which category should it hire to get maximum increase in production capacity, assuming that it needs to stich only officer's uniform :
(a) cutter (b) tailor
(c) assistant (d) can't be determined

15 If the company has to produce the shirts for only one category then of which category it can produce maximum number of uniforms?
(a) Officers
(b) Jawans
(c) either (a) or (b)
(d) none of these

Answers

✓ Level 01 *Basic Level Exercise*

1 *(c)*	2 *(b)*	3 *(c)*	4 *(a)*	5 *(d)*	6 *(b)*	7 *(b)*	8 *(c)*	9 *(b)*	10 *(c)*
11 *(b)*	12 *(c)*	13 *(a)*	14 *(a)*	15 *(b)*	16 *(b)*	17 *(a)*	18 *(a)*	19 *(b)*	20 *(a)*
21 *(a)*	22 *(c)*	23 *(d)*	24 *(c)*	25 *(b)*	26 *(b)*	27 *(a)*	28 *(c)*	29 *(c)*	30 *(c)*
31 *(b)*	32 *(a)*	33 *(b)*	34 *(d)*	35 *(a)*	36 *(c)*	37 *(c)*	38 *(b)*	39 *(b)*	40 *(d)*
41 *(a)*	42 *(b)*	43 *(b)*	44 *(c)*	45 *(b)*	46 *(a)*	47 *(a)*	48 *(c)*	49 *(a)*	50 *(a)*
51 *(c)*	52 *(a)*	53 *(d)*	54 *(b)*	55 *(a)*	56 *(c)*	57 *(b)*	58 *(a)*	59 *(d)*	60 *(a)*
61 *(a)*	62 *(c)*	63 *(b)*	64 *(d)*	65 *(c)*	66 *(b)*	67 *(b)*	68 *(a)*	69 *(b)*	70 *(b)*
71 *(b)*	72 *(c)*	73 *(d)*	74 *(a)*	75 *(c)*	76 *(d)*	77 *(b)*	78 *(b)*	79 *(b)*	80 *(c)*
81 *(b)*	82 *(a)*	83 *(a)*	84 *(a)*	85 *(b)*	86 *(a)*	87 *(b)*	88 *(d)*	89 *(c)*	90 *(c)*
91 *(d)*	92 *(c)*	93 *(a)*	94 *(b)*	95 *(a)*	96 *(c)*	97 *(a)*	98 *(b)*	99 *(c)*	100 *(a)*

✓ Level 02 *Higher Level Exercise*

1 *(a)*	2 *(c)*	3 *(c)*	4 *(b)*	5 *(c)*	6 *(c)*	7 *(c)*	8 *(b)*	9 *(c)*	10 *(d)*
11 *(d)*	12 *(b)*	13 *(c)*	14 *(b)*	15 *(c)*	16 *(d)*	17 *(b)*	18 *(c)*	19 *(c)*	20 *(d)*
21 *(a)*	22 *(c)*	23 *(b)*	24 *(c)*	25 *(b)*	26 *(d)*	27 *(b)*	28 *(d)*	29 *(c)*	30 *(b)*
31 *(c)*	32 *(d)*	33 *(b)*	34 *(c)*	35 *(a)*	36 *(c)*	37 *(d)*	38 *(d)*	39 *(c)*	40 *(b)*
41 *(c)*	42 *(b)*	43 *(b)*	44 *(a)*	45 *(c)*	46 *(c)*	47 *(d)*	48 *(d)*	49 *(b)*	50 *(c)*
51 *(c)*	52 *(c)*	53 *(b)*	54 *(b)*	55 *(c)*	56 *(c)*	57 *(c)*	58 *(d)*	59 *(c)*	

✓ Level 03 *Final Round*

1 *(c)*	2 *(b)*	3 *(a)*	4 *(c)*	5 *(c)*	6 *(c)*	7 *(a)*	8 *(b)*	9 *(d)*	10 *(b)*
11 *(c)*	12 *(b)*	13 *(d)*	14 *(b)*	15 *(c)*					

Hints & Solutions

☑ Level 01 *Basic Level Exercise*

1 Efficiency of $A = \frac{100}{12} = 8.33\%$

Efficiency of $B = \frac{100}{15} = 6.66\%$

Combined efficiency of A and $B = 8.33 + 6.66 = 15\%$

Number of days taken by A and B, when worked together

$$= \frac{100}{15} = 6\frac{10}{15} = 6\frac{2}{3} \text{ days}$$

NOTE Efficiency × Time period = Fixed amount of work

Also, in terms of percentage total work to be done is considered as 100% (in fraction it is 1), unless otherwise stated.

Alternatively

It can be done through unitary method, also.

2 Efficiency of $A = \frac{100}{10} = 10\%$

Efficiency of $B = \frac{100}{12} = 8.33\%$

Efficiency of $C = \frac{100}{15} = 6.66\%$

Combined efficiency of A, B and $C = 10 + 8.33 + 6.66$

$= 25\%$

$\therefore$ Required number of days, when A, B and C worked together $= \frac{100}{25} = 4$ days

3 Efficiency of $A = 20\%\left(= \frac{100}{5}\right)$

Efficiency of $B = 10\%\left(= \frac{100}{10}\right)$

Efficiency of A, B and $C = 50\%\left(= \frac{100}{2}\right)$

$\therefore$ Efficiency of C = (Efficiency of A, B and C) − (Efficiency of A and B)

$= (50) - (20 + 10) = 20\%$

$\therefore$ Number of days required by C to work alone

$$= \frac{100}{20} = 5 \text{ days}$$

Alternatively

Go through options and satisfy the values.

	A	B	C	$(A + B + C)$
Days →	5	10	5	2
Efficiency →	20%	10%	20%	50%

Consider option (c)

4 Efficiency of $A = 12.5\%$

Efficiency of $B = 6.25\%$

Efficiency of $C = 1.25\%$

Efficiency of $(A + B + C) = 20\% (= 12.5 + 6.25 + 1.25)$

Required number of days $= \frac{100}{20} = 5$ days

5 Efficiency of $A = 10\%$, Efficiency of $B = 6.66\%$

$\therefore$ Required number of days $= \frac{100}{16.66} = 6$ days

6 Efficiency of $A = \frac{100}{24} = 4.16\%$

Efficiency of $B = \frac{100}{30} = 3.33\%$

Efficiency of $(A + B + C) = \frac{100}{12} = 8.33\%$

$\therefore$ Efficiency of $C = (8.33) - (4.16 + 3.33) = 0.83\%$

$\therefore$ Number of days required by C to complete the work alone

$$= \frac{100}{0.83} = \frac{100}{5/6} = 120 \text{ days}$$

NOTE $\frac{5}{6} = 0.833$

7 Efficiency of $A = 10\%$ $\left\{\frac{100}{10} = 10\right.$

Efficiency of $B = 4.16\%$ $\left\{\frac{100}{24} = 4.16\right.$

Efficiency of $(A + B + C) = 16.66\%$ $\left\{\frac{100}{6} = 16.66\right.$

Efficiency of $C = (16.66) - (10 + 4.16) = 2.5\%$

Number of days required by C alone to finish the work

$$= \frac{100}{2.5} = 40 \text{ days}$$

8 Efficiency of $A = 7.14\%$

Efficiency of $B = 4.76\%$, Efficiency of $A + B = 11.9\%$

$\therefore$ Number of days required by A and B, working together

$$= \frac{100}{11.9} = 8.4 \text{ days}$$

Hint You can see that there is only one option between 8 and 9 which is 8.4 hence (c) is the correct choice.

Explanation For 8 days denominator should be 12.5 and for 9 days denominator should be almost 11.

Alternatively One day's work of A and B

$$= \frac{1}{14} + \frac{1}{21} = \frac{5}{42}$$

$\therefore$ Required number of days $= \frac{42}{5} = 8.4$ days

9 Efficiency of $A = 4.16\%$

Efficiency of $B = 1.6 \times 4.16 = 6.66\%$

$\therefore$ Number of days required by $B = \dfrac{100}{6.66} = 15$ days

Alternatively

		A	B
Efficiency	$\rightarrow$	x	$1.6\,x$
Days	$\rightarrow$	$1.6\,k$	k

(Since number of days are inversely proportional to efficiency)

Now $1.6\,k = 24 \Rightarrow k = 15$ day

Alternatively A's one day's work $= \dfrac{1}{24}$

$\therefore$ B's one day's work $= 160\%$ of $\dfrac{1}{24} = \dfrac{1}{15}$

$\therefore$ Required number of days $= 15$

10 Efficiency of $A + B = \dfrac{100}{14} = 7.14\%$

Again the ratio of efficiency of A and $B = 2:1$

$\therefore$ Efficiency of $A = \dfrac{2}{3} \times 7.14 = 4.76\%$

$\therefore$ required number of days by $A = \dfrac{100}{4.76} = 21$ days

NOTE Only choice (c) lies between 20 and 24.

Explanation For 20 days denominator should be 5 and for 24 days, denominator should be 4.16 also the only choice (c) gives a value which is multiple of 7 and very close to the answer (as appears).

Alternatively One day's work of

$$A \text{ and } B = \frac{1}{x} + \frac{1}{2x} = \frac{1}{14} \Rightarrow x = 21$$

Since A is twice efficient as B so A will take half of the days taken by B.

11 Efficiency of $A + B = 14.28\%\left(= \dfrac{100}{7}\right)$

Now, since the ratio of efficiency of A and B is $2:1$

So, the efficiency of $A = \dfrac{2}{3} \times 14.28 = 9.52\%$

$\therefore$ Number of days required by $A = 10.5$ days

Hint Only choice (b) is correct since there is no other value (in option) lies between 10 and 11.

Explanation For 10 days, denominator should be 10 and for 11 days, denominator should be 9.09.

Alternatively

Let A takes x days and then B takes $2x$ days

Then 1 day's work of A and $B = \dfrac{1}{x} + \dfrac{1}{2x} = \dfrac{1}{7}$

$\Rightarrow$ $x = 10.5$

Thus A takes 10.5 days.

12

		A		B
Efficiency	$\rightarrow$	3	:	1
Days	$\rightarrow$	1	:	3

(Since number of days are inversely proportional to the efficiency)

Now if A requires x days, so B requires $3x$ days

$\therefore$ Difference of required days $(= 3x - x) = 2x = 10 \Rightarrow x = 5$

Hence the number of days required by

$$B = 3x = 3 \times 5 = 15 \text{ days}$$

13

		A		B
Efficiency	$\rightarrow$	2	:	1
Days	$\rightarrow$	1	:	2

$$\left(\text{Days} \propto \frac{1}{\text{Efficiency}}\right)$$

Now, let A requires x days, then B requires $2x$ days

$\therefore$ difference in number of days $(= 2x - x) = x = 6 \Rightarrow x = 6$

$\therefore$ B requires $2x = 2 \times 6 = 12$ days

Alternatively If A takes x days, then B takes $x + 6$ days

Now, A's 1 day's work $= \dfrac{1}{x}$

B's 1 day's work $= \dfrac{1}{x+6}$

$\therefore$ $\dfrac{1/x}{\dfrac{1}{x+6}} = \dfrac{2}{1}$

(Since A does twice the work as B does)

$\Rightarrow$ $x = 6$

$\therefore$ B takes $2x = 12$ days.

Alternatively $\dfrac{1}{x} + \dfrac{1}{(x+6)} = \dfrac{1}{4}$

$\Rightarrow$ $x = 6$ and $2x = 12$ days (required by B)

Alternatively Go through option and satisfy all the conditions.

14

	A	:	B	:	C
Ratio of efficiency	3	:	1	:	2
Ratio of number of days	$\left\{\dfrac{1}{3}\right.$	:	$\dfrac{1}{1}$	:	$\left.\dfrac{1}{2}\right\}$
or	2	:	6	:	3

Hence, (a) is correct $\left[\because \text{Time} \propto \dfrac{1}{\text{Efficiency}}\right]$

15

	Ajit		Bablu
Efficiency	3	:	1
No. of days	1	:	3

16

	X		Y
Efficiency	5	:	1
No. of days	1	:	5

17 Efficiency of $A + B = \dfrac{100}{\dfrac{20}{3}} = 15\%$

	A		B
No. of days	4	:	5
Efficiency	5	:	4

$\therefore$ Efficiency of $B = \dfrac{4}{9} \times 15 = \dfrac{20}{3} = 6\dfrac{2}{3}\%$

18 Efficiency of $A + B = 33.33\% \left(= \frac{100}{3}\right)$

Ratio of efficiency of A and $B = 3 : 1$

$\therefore$ Efficiency of $A = \frac{3}{4} \times 33.33 = 25\%$

$\therefore$ Number of days taken by $A = 4 \left(= \frac{100}{25}\right)$

Alternatively

	A		B
Efficiency	3	:	1
No. of days	$1\,(x)$	:	$3\,(x)$

$\therefore$ Difference in days $= 2x = 8$

$\Rightarrow$ $x = 4$ and $3x = 12$

Therefore number of days taken by A, working alone $= 4$ days

Alternatively $\frac{1}{x} + \frac{1}{(x+8)} = \frac{1}{3}$

Now, you can use the options, or solve the equations to get the value of x which is equal to 4.

19 Efficiency of $(A + B) = 12.5\%$

Efficiency of $(B + C) = 8.33\%$

Efficiency of $(C + A) = 12.5\%$

$\therefore$ Efficiency of $[(A + B) + (B + C) + (C + A)]$ $= 33.33\%$

$\therefore$ Efficiency of $(A + B + C) = 16.66\%$

$\therefore$ Number of days required by A, B and C together $= 6$ days $\left(= \frac{100}{16.66}\right)$

20 Efficiency of $(A + B) = 8.33\%$

Efficiency of $(B + C) = 6.66\%$, Efficiency of $(C + A) = 5.00\%$

$\therefore$ Efficiency of $A + B + C = \frac{1}{2}(8.33 + 6.66 + 5) = 10\%$

$\therefore$ Efficiency of C = Efficiency of $[(A + B + C) - (A + B)]$ $= (10 - 8.33) = 1.66\%$

$\therefore$ Number of days required by C alone $= \frac{100}{1.66} = 60$ days

21 Remember $\text{Time} \propto \frac{1}{\text{Efficiency}}$

Now,

		Ganga	:	Jamuna	:	Saraswati
Efficiency	$\rightarrow$	3	:	1	:	2
No. of days	$\rightarrow$	2	:	6	:	3

Again, efficiency of Ganga, Jamuna and Saraswati $= \frac{100}{1} = 100\%$

$\therefore$ Efficiency of Ganga $= \frac{3}{6} \times 100 = 50\%$

Efficiency of Saraswati $= \frac{2}{6} \times 100 = 33.33\%$

Now, number of days taken by Ganga $= \frac{100}{50} = 2$

Number of days taken by Saraswati $= \frac{100}{33.33} = 3$

$\therefore$ Difference in number of days taken by Ganga and Saraswati $= 3 - 2 = 1$ day

22

Day	1	2	3	4	5	6	7	8
	A	A	A	$(A + B)$	$(A + B)$	$(A + B)$	$(A + B)$	$(A + B)$

$3 \times \frac{1}{12} = \frac{3}{12} = 25\%$ work completed

Remaining work $= 75\% = (100 - 25)$

Now, efficiency of $A + B = 15\% = (8.33 + 6.66)$

$\therefore$ Number of days required by $A + B = \frac{75}{15} = 5$ to complete rest work $(= 75\%)$

23 '3 days before the completion of the work Raja left the work' means in last 3 days only Rani has worked alone.

So, in last 3 days worked done by Rani $= 3 \times \frac{1}{21} = \frac{1}{7}$

So, the rest $\left(1 - \frac{1}{7}\right) = \frac{6}{7}$ work was done by Raja and Rani both.

Number of days in which Raja and Rani worked together $= \frac{6/7}{5/42} = \frac{36}{5} = 7\frac{1}{5}$ days

NOTE Work done by Raja and Rani in one day $= \frac{1}{14} + \frac{1}{21} = \frac{5}{42}$

$\left(\text{Also, number of days} = \frac{\text{Total work to be done}}{\text{Work to be done in one day}}\right)$

24 Work done in 11 days $= \frac{11}{30}$

Rest work $= \frac{19}{30}$

1 day's work of $A = \frac{19/30}{28} = \frac{19}{30 \times 28} = \frac{19}{840}$

Total number of days required to complete the whole work alone $= \frac{1}{19/840} = \frac{840}{19} = 44\frac{4}{19}$ days

25 Efficiency of Sonu $= 5\% = \left(\frac{100}{20}\right)$

Rest work $= 75\%$

$\therefore$ Efficiency of Abhijeet $= \frac{75}{10} = 7.5\%$

$\therefore$ Combined efficiency of Sonu and Abhijeet $= 12.5\% = (7.5 + 5)$

$\therefore$ Number of days required by Sonu and Abhijeet, to work together $= \frac{100}{12.5} = 8$ days

26

	Asha	**Usha**
Efficiency	5	4
No. of days	$4x$	$5x = 25$

$\therefore$ Number of days required by Asha to finish the work alone
$= 20 \quad \because (4x = 4 \times 5)$
(Alternatively, from percentage change graphic, number of days taken by Asha will be 20% less than Usha, if efficiency of Asha is 25% more than Usha)
Now, since Asha and Usha did work together for last 5 days
$= 5 \times 9 = 45\%$
(Since efficiency of Asha = 5% and Usha's efficiency = 4%)
It means Asha completed 55% work alone
$\therefore$ Number of days taken by Asha to complete 55% work
$= \frac{55}{5} = 11$ days

27 Krishna's efficiency = 10%, Mohan's efficiency = 5%
Work done by Krishna and Mohan together in 3 days
$= 15 \times 3 = 45\%$
Now, number of days in which B completed rest (55%) work alone $= \frac{55}{5} = 11$
Total number of days in which B worked = 3 + 11 = 14
Now number of days required by B, when A and B both worked together $= \frac{100}{15} = 6\frac{2}{3}$
$\therefore$ Required difference in number of days $= (11) - \left(6\frac{2}{3}\right)$
$= \frac{13}{3} = 4\frac{1}{3}$ days

28 Efficiency of Kareena and Karishma
$= 11.11 + 5.55 = 16.66\%$
Work done in 3 days $= 3 \times 16.66 = 50\%$
Rest work done by Kareena, Karishma and Shahid $= \frac{50}{50}$
$= 1$ day
(Since efficiency of Shahid = 33.33%)
Thus in 4 (= 3 + 1) days they have completed the work.

29 Efficiency of Kavita = 5%
Efficiency of Babita = 1.66%
Efficiency of Samita = 3.33%
Work done in 5 days by $K + B + S = 5 \times (10) = 50\%$
Work done in 3 days by $K + B = 3 \times (6.66) = 20\%$
Remaining work (30%) done by Kavita alone $= \frac{30}{5} = 6$ days

30

		A	:	B	:	C
Efficiency	$\rightarrow$	10	:	9	:	6
No. of days	$\rightarrow$	$9x$	:	$10x$	:	$15x$

Now, $15x - 9x = 6$
$\Rightarrow x = 1$
$\therefore$ Number of days taken by $A = 9$
Number of days taken by $B = 10$
Number of days taken by $C = 15$
Work done by B and C in initial 2 days $= \frac{2 \times 1}{6} = \frac{1}{3}$
rest work $= \frac{2}{3}$
Number of days required by A to finish $\frac{2}{3}$ work $= \frac{2/3}{1/9} = 6$ days

31

	Anand	Bahuguna
No. of days	45	40
Efficiency	$2.22\%\left(=\frac{1}{45}\right)$	$2.5\%\left(=\frac{1}{40}\right)$

Anand did the work in 56 days
$= 56 \times \frac{1}{45 \times 2} = \frac{28}{45}$
$\therefore$ Rest work $\left(\frac{17}{45}\right)$ was done by Anand and Bahuguna
$= \frac{17/45}{17/360} = 8$ days
(Since Anand and Bahuguna do the work in one day $= \frac{1}{45} + \frac{1}{40} = \frac{17}{360}$)

32 Efficiency of Chandni = 11.11%
Efficiency of Divakar = 8.33%
They do 19.44% work in 2 days
$\therefore$ They need 10 days to do 97.22% work,
Now the rest work (2.78) was done by Chandni in $\frac{2.78}{11.11} = \frac{1}{4}$ day
Therefore total number of days required
$= 10 + \frac{1}{4} = 10\frac{1}{4}$ days

Alternatively Chandni's one day's work $= \frac{1}{9}$
Divakar's one day's work $= \frac{1}{12}$
Chandni's and Divakar's (1 + 1) = 2 day's work
$= \frac{1}{9} + \frac{1}{12} = \frac{7}{36}$
So, in 10 days they do $= \frac{7 \times 5}{36} = \frac{35}{36}$ work
So, the remaining $\frac{1}{36}\left(=1 - \frac{35}{36}\right)$ work will be done by Chandni $= \frac{1/36}{1/9} = \frac{1}{4}$ day
Thus total number of required days $= 10 + \frac{1}{4} = 10\frac{1}{4}$ days

Hint

Day	1	2	3	4	5	6	7	8	9	10	11
Turn	C	D	C	D	C	D	C	D	C	D	C
Work	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{36}$
	7/36		7/36		7/36		7/36		7/36		
	35/36										1/36

33

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	F	Z	F	Z	F	Z	F	Z	F	Z	F	Z	F	Z
	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{12}$	$\frac{1}{60}$
	9/60		9/60		9/60		9/60		9/60		9/60		6/60	
	54/60													

In two days Fatima and Zahira do $= \frac{1}{12} + \frac{1}{15} = \frac{9}{60}$ work. In 12 days they do $= \frac{6 \times 9}{60} = \frac{54}{60}$ work

So, the remaining work $= \frac{6}{60} = \frac{1}{10}$

Now, $\frac{1}{12}$ work will be done by Fatima in 13th day

So, the remaining work $= \frac{1}{10} - \frac{1}{12} = \frac{1}{60}$

This $\frac{1}{60}$ work will be done by Zahira $= \frac{1/60}{1/15} = \frac{1}{4}$ day

Thus the total number of days required

$$= 12 + 1 + \frac{1}{4} = 13\frac{1}{4} \text{ days}$$

Alternatively A's efficiency = 8.33%, B's efficiency = 6.66%] 15%

Thus in 12 days 90% work will be done and in 13th day 8.33% more work will be done so the rest work 1.66% will be done by Zahira

$$= \frac{1.66}{6.66} = \frac{1}{4} \text{ day}$$

Thus, total number of required days $= 12 + 1 + \frac{1}{4} = 13\frac{1}{4}$

34 The difference will arise only on the last two days only. The work done on 13th day $= \frac{1}{15}$

$$\text{rest work} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$$

This rest work $\left(\frac{1}{30}\right)$ will be done by Fatima $= \frac{1/30}{1/12} = \frac{2}{5}$ day

Thus total number of required days $= 12 + 1 + \frac{2}{5} = 13\frac{2}{5}$ days.

35

1	2	3	4	5	6	7	8
A	B	C	A	B	C	A	B
1/6	1/12	1/8	1/6	1/12	1/8	1/6	1/12
3/8			3/8			1/4	

$$\frac{6}{8} = \frac{3}{4}$$

In 3 days A, B, C do $\frac{3}{8}$ work

In 6 days A, B, C do $\frac{3}{4}$ work

Rest work $= \frac{1}{4}$, which is less than $\frac{3}{8}$

On the 7th day, $\frac{1}{6}$ more work will be done by A

Now rest work $= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$

Now, this rest work $\left(\frac{1}{12}\right)$ will be done by B in 1 complete day.

Thus, total number of days = 6 + 1 + 1 = 8 days

Alternatively Efficiency of $A = 16.66\%$

Efficiency of $B = 8.33\%$

Efficiency of $C = 12.5\%$

Efficiency of $A + B = 25\%$

Efficiency of $A + B + C = 37.5\%$

In 3 days A, B, C completes 37.5% work

In 6 days A, B, C completes 75% work

Rest work = 25%

This 25% work will be completed by A and B in next 2 days

Thus total 6 + 2 = 8 days are needed.

36 From the previous solution 75% work will be completed in 6 days. In the next two days (*i.e.*, on 7th and 8th day) B and C will complete 20.83% (12.5 + 8.33) more work.

So the remaining work = 4.16%

This 4.16% work will be completed by $A = \frac{4.16}{16.66} = \frac{1}{4}$ day

So, the total number of required days $= 6 + 2 + \frac{1}{4} = 8\frac{1}{4}$ days.

37 Let B takes x days to complete the work individually.

Then, the B's 1 day's work $= \frac{1}{x}$

A's 1 day's work $= \frac{1}{x-6}$

C's 1 day's work $= \frac{1}{x-8}$

$\therefore \quad \frac{1}{x-6} + \frac{1}{x} = \frac{1}{x-8}$

Now either solve the equation or satisfy the equation from the choices given in the question.

Thus option (c) is correct.

Alternatively Select an appropriate option and then solve through % efficiency.

38 A's share = ₹ 250

B's share = ₹ 100

It means the ratio of efficiency of $A : B = 250 : 100 = 5 : 2$

$\therefore$ Ratio of days taken by A and $B = 2x : 5x$

Now, $5x - 2x = 9$

$\Rightarrow \quad x = 3$

$\therefore$ Number of days taken by $A = 6$ (efficiency = 16.66%)

Number of days taken by $B = 15$ (efficiency = 6.66%)

Therefore number of days taken by A and B, working together

$$= \frac{100}{23.33} = \frac{300}{70} = 4\frac{2}{7} \text{ days}$$

39 Alen's one day's work $= \frac{1}{21}$

Border's one day's work $= \frac{1}{42}$

(working alternatively) Alen and Border's two days work

$$= \frac{1}{21} + \frac{1}{42} = \frac{1}{14}$$

So, Alen and Border do $\frac{1}{14}$ work in 2 days

So, they complete the work in $14 \times 2 = 28$ days.

40 Efficiency of A and $B = 16.66\%$

Efficiency of B and $C = 10\%$

But efficiency of A is twice that of C

$\because$ Therefore, $A = 2C$

Now $A + B = 16.66$ and $B + C = 10$

$\Rightarrow$ $2C + B = 16.66$ …(i)

and $C + B = 10$ …(ii)

$\therefore$ from eq. (i) and (ii) $C = 6.66\%$

$\therefore$ $A = 13.33\%$

$\therefore$ Number of days taken by A to complete the work alone

$$= \frac{100}{13.33} = 7\frac{1}{2} \text{ days}$$

41 $A + B = 70\%$

$B + C = 50\%$ $\left[\because A + B + B + C - (A + B + C) = B \right.$
$\left. 70 + 50 - 100 = 20\% \right]$

$\Rightarrow$ $B = 20\%$

$A = 50\%$

and $C = 30\%$

Hence, A is most efficient.

42 $C + M = \frac{8}{17}$

$M + G = \frac{12}{17}$

$\Rightarrow$ $M = \frac{3}{17}$

$[\because (C + M + M + G) - (C + M + G) = M]$

$$\left[\left(\frac{8}{17} + \frac{12}{17}\right) - 1 = \frac{3}{17}\right]$$

$\therefore$ $C = \frac{5}{17}, G = \frac{9}{17}$

So the whole amount will be distributed in the ratio of 5 : 3 : 9 among C, M and G respectively.

Now since M is least efficient so he get his own share

$$= \frac{3}{17} \times 816 = ₹\,144$$

43

	Sharma	Kelkar
Efficiency	$0.8x$	x
Number of day	k	$0.8k = 24$

$\because$ $0.8k = 24 \Rightarrow k = 30$

Thus Sharma requires 30 days, to complete the work, alone.

44

	Man	Days	
$\frac{1}{5}\uparrow$	30	24	$\frac{1}{6}\downarrow$
	36	20	

Applying product constancy method.

For a constant work when days are reduced by $\frac{1}{6}$, then number of men is increased by $\frac{1}{5}$. Hence 6 men will increase.

Alternatively $24 \times 30 = 20 \times x$

$\Rightarrow$ $x = 36$

Therefore 6 more men are required.

45

	Women	Days	
$\downarrow\frac{1}{3}$	12	20	$\frac{1}{2}\uparrow$
	8	30	

Thus 10 more days are required.

Alternatively $12 \times 20 = 8 \times x$

$\Rightarrow$ $x = 30$

$\therefore$ 10 more days are required.

46

	Boys	Days	
$\frac{2}{5}\downarrow$	35	15	$\frac{2}{3}\uparrow$
	21	25	

Thus 14 boys did not turn up for the job.

Alternatively $35 \times 15 = 25 \times x$

$\Rightarrow$ $x = 21$

$\therefore$ $35 - 21 = 14$ boys did not turn up for the job.

47 $x \times 32 = 24 \times 40$

$x = 30$

$(M_1D_1 = M_2D_2)$

48 $M \times D = W$

$16 \times 6 = \frac{1}{3}W$

Rest work $= \frac{2}{3}W$

For double work in same time we need double men. So 16 more men are required.

Alternatively $2 \times (16 \times 6) = 6 \times M$

$\Rightarrow$ $M = 32$

$\therefore$ 16 more men are required.

49 $M \times D = 10 \times 20 = 200$ Man-days

New Man-days $= (20 \times 2) \times x$

$200 = 20 \times 2 \times x$

$x = 5$ days

or $M_1D_1 = M_2D_2$

$10 \times 20 = (20 \times 2) \times x$

$\Rightarrow$ $x = 5$

50 $M_1D_1 = M_2D_2$

$M_1 \times 20 = (M_1 - 12) \times 32$

$\Rightarrow$ $M_1 = 32$

Also, using the above concept you can go through options.

51 Go through options. Consider option (c)

$$30 \times 20 = 30 \times 6 + 21 \times 20$$

$600 = 600$, hence presumed option is correct.

Alternatively $30 \times 20 = 30 \times x + 21 \times (26 - x)$

$$\Rightarrow \quad x = 6$$

52 Go through options. Consider choice (a)

$25 \times 30 = 25 \times 15 + 75 \times 5$

$750 = 750$, hence choice (a) is correct.

Alternatively $25 \times 30 = 25 \times x + 75 \times (20 - x)$

$$\left[\because 30 \times \frac{2}{3} = 20\right]$$

53 Go through options. Consider choice (d)

$$36 \times 120 = 40 \times 108$$

$$4320 = 4320$$

Hence, choice (d) is correct.

Alternatively

	Men	Days	
$\frac{1}{9}\uparrow$	x	120	$\left(\frac{1}{10}\right)\downarrow$
	$\downarrow$	$\downarrow$	
	$\left(x \times \frac{10}{9}\right)$	(12)	

From percentage change (product constancy) graphic when number of days are decreased by $\frac{1}{10}$ then the number of men are increased by $\frac{1}{9}$ and $\frac{1}{9}$ is equivalent to 4 men so the actual number of men are $9 \times 4 = 36$.

Alternatively $M_1 \; D_1 = M_2 \quad D_2$

$$x \times 120 = (x + 4) \times 108$$

$$\Rightarrow \quad x = 36$$

54 Efficiency (per minute) of Modi = 4 copies/min

Efficiency of Modi and Xerox together = 10 pages/min

$\therefore$ Efficiency of Xerox alone = $10 - 4 = 6$ pages/min

$\therefore$ Mr. Xerox needs 6 minutes to copy 36 pages.

55

$$\text{Work done} = \frac{1}{3}$$

$$\text{Remaining work} = \frac{2}{3}$$

$$2 \times (20 \times 12) = 12 \times x$$

$$\Rightarrow \quad x = 40$$

So, 20 men will be increased.

56

$$\text{Work done} = \frac{1}{5}$$

$$\text{remaining work} = \frac{4}{5}$$

$$\therefore \quad 4\,(20 \times 75) = 40 \times x$$

$$x = 150$$

Therefore 75 men should be increased.

57 Work done by 6 men = Work done by 10 women.

$\Rightarrow$ Work done by 1 man = work done by $\frac{10}{6} = \frac{5}{3}$ women

$$\therefore \quad 12 \text{ men} + 5 \text{ women} = 12 \times \frac{5}{3} + 5 = 25 \text{ women}$$

$$\therefore \quad W_1 \times D_1 = W_2 \times D_2, W = \text{women}, D = \text{days}$$

$$10 \times 15 = 25 \times D_2 \Rightarrow D_2 = 6$$

58 Work done by 2 men = 3 women = 4 boys

$$\therefore \quad 1 \text{ man} = 2 \text{ boys}$$

$$1 \text{ woman} = \frac{4}{3} \text{ boys}$$

$$\therefore \quad \text{boys} \times \text{days} = 4 \times 52 \text{ (boys-days)}$$

Again 1 man + 1 woman + 1 boy $= 2 + \frac{4}{3} + 1 = \frac{13}{3}$ boys

$$B_1 \times D_1 = B_2 \times D_2, \quad B = \text{boys}, D = \text{days}$$

$$4 \times 52 = \frac{13}{3} \times D_2$$

$$D_2 = 48 \text{ days}$$

59 2 men = 7 boys $\Rightarrow$ 1 man $= \frac{7}{2}$ days

5 women = 7 boys $\Rightarrow$ 1 woman $= \frac{7}{5}$ boys

7 men + 5 women + 2 boys $= 7 \times \frac{7}{2} + 5 \times \frac{7}{5} + 2 = \frac{67}{2}$ boys

Now, $B_1 \times D_1 = B_2 \times D_2$

$$7 \times 469 = \frac{67}{2} \times D_2$$

$$\Rightarrow \quad D_2 = 98 \text{ days}$$

60

$$6C + 2M = 6 \text{ days}$$

$$\Rightarrow \quad 36C + 12M = 1 \text{ days}$$

Again $1M = 2C$

$$\therefore \quad 36 + 12 \times 2 = 1 \text{ day}$$

60 children can do the work in 1 day

Now, 5 men = 10 children

$\therefore$ 10 children can do the work in 6 days.

61

$$8M + 12W = 4 \text{ days (whole work)}$$

$$\Rightarrow \quad 32M + 48W = 1 \text{ day} \qquad \ldots(i)$$

Again $6M + 14W = 5$ days

$$\Rightarrow \quad 30M + 70W = 1 \text{ day} \qquad \ldots(ii)$$

From eq. (i) and (ii)

$$32M + 48W = 30M + 70W$$

$$\Rightarrow \quad 2M = 22W$$

$$\Rightarrow \quad 1M = 11W$$

Now, $30M + 70W = 1$ day

$$(30 \times 11 + 70)W = 1 \text{ day}$$

Therefore $400W$ requires 1 day to complete the whole work.

Thus $20W$ needs 20 days to complete the whole work.

62 Efficiency of $A = 6.66\%$

Efficiency of $B = 5.55\%$

Efficiency of $A + B + C = 16.66\%$

$\therefore$ Efficiency of $C = 4.44\%$

Now, number of days required by

$$C = \frac{100}{4.44} = \frac{100}{(10 \times 4)/9} = \frac{45}{2} \text{ days}$$

63 Ratio of efficiencies of A, B and $C = 6 : 5 : 4$

$\therefore$ Share of $C = \frac{4}{15} \times 27000 = 7200$

64 In 1 hour 314 weavers weave $= 6594 \times 6$ shawls

In 1 hour 1 weaver weaves $= \frac{6594 \times 6}{314}$ shawls

$= 126$ shawls

65 $3M + 2W = 4$ days

$\Rightarrow \quad 12M + 8W = 1$ day ...(i)

Again, $2M + 3W = 5D$

$\Rightarrow \quad 10M + 15W = 1D$ ...(ii)

From eq. (i) and (ii)

$12M + 8W = 10M + 15W$

$\Rightarrow \quad 2M = 7W$

$\Rightarrow \quad \frac{M}{W} = \frac{7}{2}$

Since the ratio of efficiency of women : men $= 2 : 7$

So, the amount of a man per day $= ₹\,154$

$\left[\left(\frac{7}{2} \times 44\right) = 154\right]$

66 Go through options.

Alternatively

$30 \times 40 = 30 \times x + 20 \times (46 - x) \Rightarrow x = 28$ days

67 Women $\times$ hours $= 8 \times 10 = 80$ women hours

Now, women $\times$ hours $= 12 \times 8 = 96$ women hours

Since new work force is 20% greater than previous work force.

So, the new quantity of tea leaves will be increased by 20% which is equal to $200 \times \frac{20}{100} = 40$ kg, hence (b)

68 New work $= 3 \times 450$ man-day

$\therefore \quad 3 \times 450 = 27 \times x$

$x = 50$ days

69 $4B + 5G = 10$

$\Rightarrow \quad 40B + 50G = 1$ ...(i)

Again, $6B + 6G = 7$

$\Rightarrow \quad 42B + 42G = 1$ ...(ii)

Comparing eq. (i) and (ii)

$40B + 50G = 42B + 42G$

$\Rightarrow \quad 2B = 8G \quad \Rightarrow \quad 1B = 4G$

Now, $(42 \times 4 + 42)$ girls $= 1$ day

210 girls can do a work in 1 day

Again $2B + 7G = 15$ girls

So, 15 girls require $\frac{210}{15} = 14$ days

70 Equate the man-days

For 20 km road, $20 \times 20 = 400$ man-days are required

$\therefore$ For 40 km road 800 man-days are required

So, $800 = 40 \times x \Rightarrow x = 20$ days

71 $\frac{3}{4} \times (x - 2)\, x = (x + 7)(x - 10)$

$\Rightarrow \quad x^2 - 6x - 280 = 0$

$\Rightarrow \quad x = 20$ and $x = -14$

So, the acceptable values is $x = 20$

$\therefore$ Total work $= (x - 2) \times x = 18 \times 20 = 360$ unit

Now $360 = 30 \times k \quad \because (30 = 20 + 10)$

$\Rightarrow \quad k = 12$ days

72 Efficiency of a man : woman : girl $= 6 : 3 : 1$

$\therefore$ Share of a woman and girl $= \frac{(3 + 1)}{(6 + 3 + 1)} \times 10{,}000$

$= \frac{4}{10} \times 10000 = ₹\,4000$

73 Total work $= 33 \times 30 = 990$ man-days

First day's work $= \frac{1}{990} \times 44$

Second day's work $= \frac{1}{990} \times 43$

Third day's work $= \frac{1}{990} \times 42$ and so on

So, the total work in 44 days $= \frac{1}{990}(44 + 43 + 42 + \ldots)$

$= \frac{1}{990} \times \frac{44 \times 45}{2} = 1$

Hence in 44 days total work will be completed.

74 Efficiency of Abhishek $= 2.5\% = \left(\frac{100}{40}\right)$

Work done in 8 days $= 20\%$

Rest work $= 80\%$

Efficiency of Bacchhan $= \frac{80}{24} = 3.33\%$

$\therefore$ Required number of days $= \frac{100}{(2.5 + 3.33)}$

$= \frac{100}{5.83} = \frac{100 \times 6}{35} = 17\,\frac{1}{7}$

Hint $0.83 = \frac{5}{6}$, so, $5.83 = 5 + \frac{5}{6} = \frac{35}{6}$

75

	A		B		C
Efficiency	3	:	2	:	6
Number of days	2	:	3	:	1

$\therefore$ Number of days taken by $A = 12$

Number of days taken by $B = 18$

Number of days taken by $C = 6$

1 day's work of $(A + B) = \frac{5}{36}$

1 day's work of $(B + C) = \frac{8}{36}$

1 day's work of $(C + A) = \frac{9}{36}$

Day	→	1	2	3	4	5	6
Work	→	5/36	8/36	9/36	5/36	8/36	
				35/36			1/36

In 5 days total work done $= \frac{35}{36}$

Now, the rest work (1/36), which is done by AC.

$\therefore$ Number of days taken by AC for the rest work

$$= \frac{1/36}{9/36} = \frac{1}{9}$$

Therefore, total time $= 5 + \frac{1}{9} = 5\frac{1}{9}$ days

76 Efficiency of $A + B = 16.66$, Efficiency of $B + J = 10$

We have no further relevent informations, so we cannot determine.

77

	A	B
Efficiency	x	$2x$
Number of days	15	15/2

A's one day's work $= \frac{1}{15}$, B's one day's work $= \frac{2}{15}$

$$(A + B)\text{'s one day's work} = \frac{3}{15}$$

Now, let us assume B joined A after $(11 - x)$ days, then

$$\frac{(11-x)}{15} + \frac{x \times 3}{15} = 1 \Rightarrow (11 - x) + 3x = 15 \Rightarrow x = 2$$

It means they worked together for 2 days.

Alternatively Go through options and check easily with the percentage efficiency.

$$9 \times 6.66 + 2 \times 20 = 100\%$$

78 Efficiency of $A = 6.66\%$

Efficiency of $B = 3.33\%$

Efficiency of $C = 2.5\%$

Work done in last two days (only C do it) $= 2 \times 2.5 = 5\%$

Work done in the 3rd and 4th day from the last day (only A and C do it) $= 2 \times 9.16 = 18.33\%$

Remaining work $= 100 - (5 + 18.33) = 76.66\%$

This 76.66% work was done by all of A, B and C.

$\therefore$ Number of days taken by them $= \frac{76.66}{12.5} = \frac{460}{75} = 6\frac{2}{15}$

$\therefore$ Total time required $= 6\frac{2}{15} + 2 + 2 = 10\frac{2}{15}$ days

79 Let x kg of oil is used for eating purpose, daily, then

$$(x + 11) \times 50 = (x + 15) \times 45$$

$$x = 25$$

$\therefore$ Total quantity of oil $= (25 + 11) \times 50 = 1800$

$\therefore$ Required number of days $= \frac{1800}{25} = 72$ days

80 Work done $= \frac{2}{3}$

Remaining work $= \frac{1}{3}$, which is half of $\frac{2}{3}$

$\therefore \frac{1}{2} \times (20 \times 32) = 8 \times x \Rightarrow x = 40$ men

Therefore, 20 more men were required.

81 One day's work of 7 Indian with 4 Chinese $= \frac{1}{5}$

and one day's work of 7 Japanese and 3 Chinese $= \frac{1}{7}$

Therefore, one day's work of 7 Indian, 7 Chinese and 7 Japanese $= \frac{1}{5} + \frac{1}{7} = \frac{12}{35}$

Therefore, one day's work of 1 Indian, 1 Chinese and 1 Japanese $= \frac{12}{35} \times \frac{1}{7} = \frac{12}{35 \times 7}$

Therefore, number of days required by 1 Indian, 1 Chinese and 1 Japanese

$$= \frac{1}{12/(35 \times 7)} = \frac{35 \times 7}{12} = 20\frac{5}{12} \text{ days}$$

82 In 1 minute A, B and C bind $= \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48}$ book

In 12 hours A, B and C bind $= 12 \times 60 \times \frac{13}{48} = 195$ books

$\therefore$ Average number of books bind by each $= \frac{195}{3} = 65$ books

83 Work of a man for 1 hour $= \frac{3}{2}$ women's work for 1 hour

Again, work of a man for 1 day

$$= \left(\frac{3}{2} \times \frac{9}{7.5}\right) \text{ women's work for 1 hour}$$

$\Rightarrow$ Work of a man for 1 day $= \frac{9}{5}$ women's work for 1 day

$\Rightarrow$ 1 man $= \frac{9}{5}$ women

$\therefore$ 10 men + 6 women $= 10 \times \frac{9}{5} + 6 = 24$ women

$\therefore$ 10 men + 9 women $= 10 \times \frac{9}{5} + 9 = 27$ women

Now, $D_1 \times W_1 = D_2 \times W_2$

$$18 \times 24 = D_2 \times 27 \Rightarrow D_2 = 16 \text{ days}$$

84 Number of days taken by A to complete work alone $= 14$ days

Number of days taken by B to complete work alone $= 7$ days

Number of days taken by C to complete work alone $= 7$ days

One day's work of A and $B = \frac{1}{14} + \frac{1}{7} = \frac{3}{14}$

and one day's work of A, B and $C = \frac{1}{14} + \frac{1}{7} + \frac{1}{7} = \frac{5}{14}$

3 day's work of A and $B = 3 \times \frac{3}{14} = \frac{9}{14}$

remaining work $= \frac{5}{14}$ $\left(1 - \frac{9}{14}\right)$

This remaining work will be done by A, B and C

$$= \frac{5/14}{5/14} = 1 \text{ day}$$

85 A's 5 days work = 50%

B's 5 days work = 33.33%

C's 2 days work = 16.66% [100 – (50 + 33.33)]

Ratio of contribution of work of A, B and $C = 50 : 33\frac{1}{3} : 16\frac{2}{3} = 3 : 2 : 1$

A's total share = ₹ 1500

B's total share = ₹ 1000

C's total share = ₹ 500

A's one day's earning = ₹ 300

B's one day's earning = ₹ 200

and C's one day's earning = ₹ 250

86 Efficiency of $A = 50\%$ $\left(\frac{100}{2}\right)$

Efficiency of $B = 37.5\%$ $\left(\frac{100}{8/3}\right)$

Efficiency of $C = 12.5\%$ $\left(\frac{100}{8}\right)$

Combined efficiency of A, B and $C = 100\%$

So, they complete the work in 1 hour.

87

	A	+	B	=	C	+	D
	↓		↓		↓		↓
Ratio of efficiency	$\underbrace{10x}_{15x}$	+	$\underbrace{5x}_{15x}$		$\underbrace{9x}_{15x}$	+	$\underbrace{6x}_{15x}$

Therefore, ratio of efficiency of $A : C = 10 : 9$

Therefore, ratio of days taken by $A : C = 9 : 10$

Therefore, number of days taken by $A = 18$ days

88 Efficiency of 4 men and 2 boys = 20%

Efficiency of 3 women and 4 boys = 20%

Efficiency of 2 men and 3 women = 20%

∴ Efficiency of 6 men, 6 women and 6 boys = 60%

∴ Efficiency of 1 man, 1 woman and 1 boy = 10%

Now, since they will work at double their efficiency

∴ Efficiency of 1 man, 1 woman and 1 boy = 20%

∴ Required number of days = 5

89 $M_1 \times D_1 = M_2 \times D_2$

$m \times r = (m + n) \times D_2$

$\Rightarrow \quad D_2 = \frac{mr}{(m+n)}$

90 Efficiency of $A = 2.77\%$, Efficiency of $B = 2.22\%$

∴ Combined efficiency of A and $B = 5\% = (2.77 + 2.22)$

Thus, it will take total of 20 minutes $\left(\because 20 = \frac{100}{5}\right)$

91 Efficiency of $A = 8.33\%$

Effective efficiency = 6.66%, when there is leakage

∴ Efficiency of leakage = 1.66% = (8.33 – 6.66)

It means due to leakage a full tank will be empty in 60 hours.

92 Efficiency of $A + B = 10 + 6.66 = 16.66\%$

Efficiency of $A + B + C = 5.55\%$

∴ Efficiency of C (outlet pipe) = 16.66 – 5.55 = 11.11%

It means outlet pipe C can empty in 9 hours.

93 Efficiency of only leakage = 16.66%

Effective efficiency of leakage = 6.66%

It means the capacity of filling pipe = 10%

Therefore, the inlet pipe can fill the tank in 10 hours hence the capacity of tank = 100 l.

94 Efficiency of tap A and $B = 16.66\% = (10 + 6.66)$

$\therefore \quad 16.66x + 10 \times (8 - x) = 100\% \Rightarrow x = 3$

95 Efficiency of $A = 5\%$, Efficiency of $B = 4\%$

Efficiency of $C = -3.33\%$

It means in every 3 consecutive hours tops A, B and C can fill 5.66% (= 5 + 4 – 3.33). Therefore in 51 hours (= 3 × 17) tops A, B and C can fill 96.33% (= 5.66 × 17)

∴ the remaining part *i.e.,* 3.66% (= 100 – 96.33) can be filled up by A in $\frac{11}{15}$ hours $\left(= \frac{3.66}{5}\right)$, since it is now A's turn.

Hence, the total time required = $51 + \frac{11}{15} = 51\frac{11}{15}$

97 Efficiency when both pipes used to fill = $A + B$

and efficiency when pipe A is used to fill and pipe B is used to empty the tank = $A - B$

$\therefore \frac{A+B}{A-B} = \frac{2}{1} \Rightarrow \frac{A}{B} = \frac{3}{1}$ (By componendo and dividendo)

Thus, the ratio of efficiency of pipe A and $B = 3 : 1$.

98 **In ideal case:**

Time taken to fill the half tank by A and $B = \frac{50}{41.66} = \frac{6}{5}$ hours

Time taken by A, B and C to fill rest half of the tank

$= \frac{50}{16.66} = 3$ hours

Total time $= \frac{6}{5} + 3 = 4$ hours 12 minutes

In second case:

Time taken to fill $\frac{3}{4}$ tank by A and $B = \frac{75}{41.66} = \frac{9}{5}$ hours

Time taken by A, B and C to fill rest $\frac{1}{4}$ tank $= \frac{25}{16.66} = \frac{3}{2}$ hours

Total time $= \frac{9}{5} + \frac{3}{2} = 3$ hours 18 minutes

Therefore, difference in time = 54 minutes

99 Time taken by pipes A and B to fill the whole tank

$= \frac{100}{16.66} = 6$ hours

Capacity filled in 2 hours by pipes A, B and $C = 2 \times 13.33$

$= 26.66\%$

Remaining capacity = 73.33%

This remaining capacity can be filled by A and B

$= \frac{73.33}{16.66} = 4\frac{2}{5}$ hours

So, the total time required = $2 + 4\frac{2}{5} = 6$ hours 24 minutes

Thus, in this case 24 minutes extra are required.

100 In one hour pipe A can fill $= \frac{1}{30}$ part of the tank

In one hour pipe B can fill $= \frac{1}{45}$ part of the tank

In two hour pipes A and B can fill $= \frac{1}{18}$ part of the tank

Therefore in 36 hours the tank will be completely filled.

Alternatively Efficiency of pipe $A = 3.33\%$

Efficiency of pipe $B = 2.22\%$

and Combined efficiency $= 5.55\%$

Therefore in 2 hours pipe A and B fill 5.55%.

Thus to fill 100% tank, these pipe will take 36 hours.

☑ Level 02 *Higher Level Exercise*

1 LCM of 2, 3 and 4 = 12

In 12 hours A will make 6 shawls
" " " B will make 4 shawls
" " " C will make 3 shawls
} 13 shawls

i.e., in 12 hours they will weave 13 shawls

so, in 84 hours they will weave 91 shawls

Now, in 9 hours A will make 4 shawls
in 9 hours B will make 3 shawls
in 9 hours C will make 2 shawls
} 9 shawls

So, they will complete 100 shawls in 93 hours.

NOTE Since, they cannot share each-others work so B will take completely 9 hours to make 3 shawls, even when A and C stay idle for the last 1 hour till B completes his own work.

2

	M T	W Th	F S
	A S	A S	A S
Work →	15%	15%	7.5%
	37.5%		

{This pattern continued for total 2 weeks only till 75% work got completed.}

Thus in 2 weeks they will complete 75% work.

Now 15% of the remaining (25% of the work) will be done in the third week in Monday and Tuesday. Again 10% work remained undone. Out of this 8.33% work will be done by Arun on Wednesday and remaining 1.66% work will be completed on Thursday by Satyam.

Final week

M T	W	Th
A S	A	S
15%	8.33%	1.66%
25%		

3 Efficiency of Kaushalya = 5%

Efficiency of Kaikeyi = 4%

Thus, in 10 days working together they will complete only 90% of the work.

$$[(5 + 4) \times 10] = 90$$

Hence, the remaining work will surely done by Sumitra, which is 10%.

Thus, Sumitra will get 10% of ₹ 700, which is ₹ 70.

Solutions (for Q. Nos. 4 and 5)

	1st day	2nd day
Morning shift}	A	B
Evening shift}	B	A

It is clear that in two days finally they work very similar to the alternate days *i.e.*, finally A work for 10 hours and B also works for 10 hours. Thus in every two days they will complete 7.5% work. So, in 26 days they will complete 97.5% of the total work. Now, the remaining work = 2.5%

Now, this is the turn of A,

Since A does 4.16% work in 10 hours.

So, he will do 2.5% work in 6 hours.

4 Thus, the work will be finished on 27th day.

5 Since, A does 2.5% work in 6 hours, which is the actual duration of morning shift. So 100% time of the morning shift was utilised.

6 **From the first statement :**

	$(B + C)$	$(A + B + C)$
Efficiency	$2x$	$3x$
Days	$3y$	$2y$

Thus, we can say that efficiency of A is $\frac{1}{3}$ the efficiency of $(A + B + C)$.

From the last statement:

$$\text{Share of } B \text{ out of total amount} = \frac{120}{450} = \frac{4}{15}$$

From these two results we can conclude that:

		A	:	B	:	C
Ratio of efficiency		$\frac{5}{15}\left(=\frac{1}{3}\right)$	:	$\frac{4}{15}$	:	$\frac{6}{15}$
	$\Rightarrow$	5	:	4	:	6
Ratio of number of days	$\Rightarrow$	$\frac{1}{5}$	:	$\frac{1}{4}$	:	$\frac{1}{6}$
	$\Rightarrow$	$12x$	:	$15x$	:	$10x$

$$\text{One day's work of } A \text{ and } B = \frac{1}{12x} + \frac{1}{15x} = \frac{9}{60x}$$

$\therefore$ A and B will take $\frac{60x}{9}$ days to complete the whole work,

Again one day's work of A, B and C

$$= \frac{1}{12x} + \frac{1}{15x} + \frac{1}{10x} = \frac{15}{60x}$$